



Asset Performance Management

Data Concentrators

User Guide

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ABOUT THIS GUIDE

The Data Concentrator is a file collection service that allows you to collect data from multiple systems, and pass it into your Historians or Honeywell's Asset Performance Management.

This manual outlines the procedures and guidelines to configure the Data Concentrator for particular requirements.

PRE-REQUISITES

Below are the requirements for the Data Concentrator to function:

1. Microsoft .NET Framework v3.5
2. Microsoft .NET Framework v4.5.2
3. Access to a Microsoft SQL Server database instance.
4. Access to one or more historians or an asset manager server to which process data will be written.
5. Access to an Alarm Manager TCP/IP Collector Channel, preferably on the local machine.
6. If you are using SQL CE for the tracking database.
 - Install SQL Server Compact Edition 3.5 SP2 on the computer where the Data Concentrator will be installed.

NOTE

If you are using a 64-bit system, install the 32-bit version first, and then install the 64-bit version.

7. If you are writing data to OPC:
 - ASP .Net 4.51
 - Core OPC libraries on the Data Concentrator server.
8. If you are writing data to PHD:
 - PHD R320+ Client on the Data Concentrator server for writing to PHD. The x86 Client should be used for x86 Data Concentrator and the x64 Client should be used for x64 Data Concentrator.
9. If you are using SFTP to transfer files from the Data Concentrator Forwarding Agent:
 - Access to SFTP server, to which files will be forwarded after processing.

2.1 Core OPC Libraries (Optional)

If no OPC servers have been installed on the Data Concentrator machine, it is necessary to install the Core OPC Libraries. Follow the below steps to install OPC servers:

1. Navigate to **Prerequisites** folder in the installation media and extract the content from below zip file.
OPC Core Components Redistributable 3.00.102.zip
2. Execute **OPC Core Components Redistributable.msi**, accepting all defaults.

INSTALLATION

3.1 Preparing Data Concentrator Database

Follow the below steps in Database Server before running Data Concentrator installation wizard to use with Asset Performance Management.

1. Navigate to the below folder in the server with Asset Performance Management database.
\\Utilities\\DataConcentrator\\DataConcentrator\\SQL Scripts\\
2. Run **UAS_ConfigureDataConcentratorDB.ps 1** as an administrator in powershell.
3. When prompted, provide the following details:
 - a. The SQL server instance name (**LocalSQLServerInstanceName**) in the below format:
[SQL_NAME]\\[INSTANCE_NAME]

NOTE

For default SQL instance, use only the machine name.

- b. The Asset Performance Management services user name (**ServiceUserLoginName**) in the below format:
[DOMAIN_NAME]\\[USER_NAME]

3.2 Installing the Data Concentrator Service

Follow the below steps in Application Server to install the Data Concentrator Service:

1. Navigate to the following folder in the installation media and double-click the **Honeywell APM Data Concentrator.msi** file to open the installation wizard.
<Installation Media>:\\Utilities\\DataConcentrator\\DataConcentrator\\

NOTE

For Windows 2012, the user with permission to create directories should run the installer.

2. Select **Data Concentrator Installation**, and click **Next**.
3. Click **Install**, and follow the steps in the **Data Concentrator Setup Wizard**.
4. Click **Close** after the installation is completed.
5. Open **Services** from Control Panel (Control Panel > Administrative Tools > Services).
6. Navigate to the following folder and open **Matrikon.Datalogger.Collector.Service.exe.config** to update the connectionString to the point of your Server instance:

<install Location>:\Program Files
(x86)\Honeywell\MES\AssetManager\DataConcentrator\DataConcentrator\Matrikon.Datalogger.Collector.Service
.exe.config

7. If APM server is using https protocol, then in the data concentrator configuration file change the Security Mode under basicHttpBinding node to Transport as given below:

```
<bindings>
<basicHttpBinding>
<binding name="BasicHttpBinding_Common">
<security mode="Transport">
<transport clientCredentialType="Windows" />
</security>
</binding>
</basicHttpBinding>
</bindings>
```

8. Save and close the configuration file.
9. In the Services panel, right-click on **Honeywell APM Data Concentrator** and select **Properties**.
10. Change the service's user to the Asset Performance Management services user.
11. Configure the Data Concentrator as described in the [Data Concentrator Configuration](#).

3.3 Completing Configuration for APM Autosync

NOTE
If APM is using the https protocol, ensure the setting for the url under **System > System Configuration > Application** is using https as in the following example:
`https://<webservername.fullyqualified domain name:portnumber>/MES/AM`

NOTE
DC to APM auto synchronization is set to **False** by default in MES application page. If you want to enable it, set it to **True**.

System > System Configuration > Application

Key	Value	Owner
Honeywell UAS Data DC.DataSourceNames	Data Concentrator	UAS
Honeywell UAS Data DC.DefaultDataSourceConnectionInformation	<ConnectionXml><Type>DC</Type><instanceName>DC1</instanceName><DataloggerTemplateConfigurations><...	UAS
Honeywell UAS Data DC.DefaultDataSourceName	Data Concentrator	UAS
Honeywell UAS Data DC.EnableUOMConversions	True	UAS
Honeywell UAS Data DC.RemoveTagsNotDefinedInUAS	True	UAS
Honeywell UAS Data DC.StateRefreshMaxRetries	10	UAS
Honeywell UAS Data DC.StateRefreshSeconds	30	UAS
Honeywell UAS Data DC.SyncConfigFromUAS	False	UAS
Honeywell UAS Data DC.SyncFromUAS	True	UAS
Honeywell UAS Data DC.LUTC	True	UAS

Follow the below steps to configure APM Autosync:

1. Open command prompt as administrator in the APM machine.
2. Enter **certlm** in the command prompt and press enter.
The **Microsoft Management Console** page is displayed.
3. Navigate to Personal folder and Export the certificate used by Rabbit MQ.

NOTE

Export it with password.

4. Copy the certificate in Data Concentrator machine.
5. Navigate to the following folder in installation media and open a command prompt:
Utilities\DataConcentrator\DataConcentrator\WorrakerConfiguration
6. Run the **WorkerConfiguration** using the following command (updated to include the required details):

```
WorkerConfiguration.exe -c [path to DC certificate] -p [password of DC certificate file] -u [DC service account name of the native machine] -s [protocol]://[UASWebServer]:[port]
```

NOTE

This requires the location and password of the RabbitMQ certificates generated in step 1.

- a. If you have a simple password (example:Password1) refer the below example.

For example:

```
WorkerConfiguration.exe -c C:/RabbitMQCert -p Password1 -u ourdomain\UASServices -shttps://UASWebServer.ourdomain.com:443
```

- b. If you have a complex Password (ex:Password@1) enclose the password with "", refer the below example.

For example:

```
WorkerConfiguration.exe -c C:/RabbitMQCert -p Password@1 -u ourdomain\UASServices -s https://UASWebServer.ourdomain.com:443
```

NOTE

If you install Data Concentrator in a separate node (e.g on FDM node), DOT Net 4.7.1 is required for configuring the RabbitMQ certificate. Dot Net 4.7.1 is available in the below location:

<Install Media>\Prerequisites\NDP471-KB4033342-x86-x64-AllOS-ENU.exe.

NOTE

If Data Concentrator is standalone, then copy the Rabbit MQ issuing authority from APM machine and paste it in the Data Concentrator machine under trusted root authority.

7. Ensure the Data Concentrator service is started.

3.4 Installing the Data Concentrator Forwarding Agent Service (Optional)

Follow the below steps in Application Server to install the Data Concentrator Forwarding Agent Service:

1. Navigate to the following folder in the installation media and double-click the **Honeywell APM Forwarding Agent.msi** file to open the installation wizard:

<Installation Media>:\Utilities\DataConcentrator\DataConcentrator\

NOTE

For Windows 2012, the user with permission to create directories should run the installer.

2. Select **Forwarding Agent Installation**, and click **Next**.
3. Click **Install**, and follow the steps in the **Data Concentrator Forwarding Agent Setup Wizard**.
4. Click **Close** after the installation is completed.
5. Configure the Forwarding Agent Service as described in [Data Concentrator Forwarding Agent Service](#).

3.5 Installing the Data Concentrator Archive Maintainer (Optional)

Follow the below steps in Application Server to install the Data Concentrator Archive Maintenance Service:

1. Navigate to the following folder in the installation media and double-click the **Honeywell APM Archive Maintainer.msi** file to open the installation wizard:

<Installation Media>:\Utilities\DataConcentrator\DataConcentrator\

NOTE

For Windows 2012, the user with permission to create directories should run the installer.

2. Select **Data Concentrator Archive Maintainer**, and click **Next**.
3. Click **Install**, and follow the steps in the **Data Concentrator Archive Maintainer Setup Wizard**.
4. Click **Close** after the installation is completed.
5. Configure the Archive Maintainer as described in [Data Concentrator Archive Maintainer Configuration \(Optional\)](#).
6. Create a Windows Scheduled Task to run **Matrikon.Datalogger.ArchiveMaintainer.exe** at the desired maintenance intervals.

4.1 Upgrading the Data Concentrator

4.1.1 Prerequisites

Following are the requirements to upgrade the Data Concentrator:

1. Data Concentrator should be correctly installed and configured.
2. User with administrative privileges to perform the install.
3. Installation media.

4.1.2 Systems with Existing APM Configuration

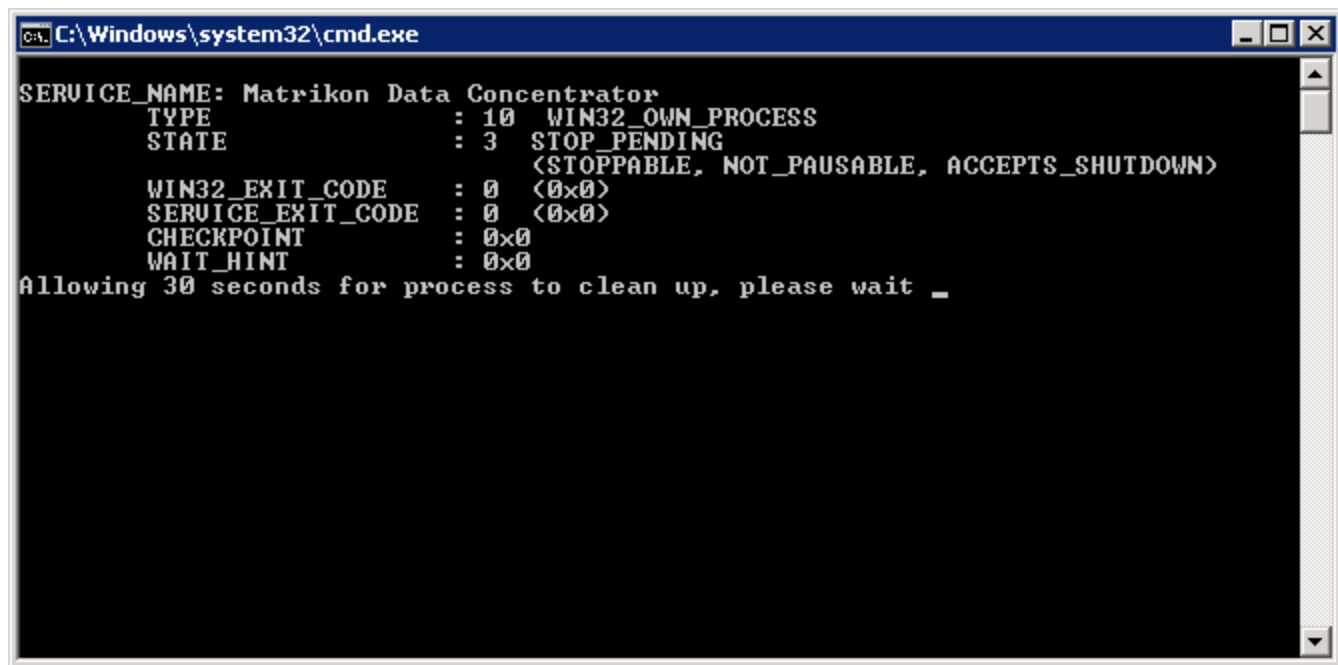
If you are updating a system with existing connections to Asset Performance Management and has existing tag configuration, the data should be migrated to ensure it is not lost during the APM Autosync starts up. For information on migrating your data, please contact Honeywell Customer Support.

4.1.3 Upgrade Procedure

Follow the below steps to upgrade the Data Concentrator:

1. Login as an administrative user in the computer with Data Concentrator.
2. Navigate to the following folder and take a backup of Data Concentrator configuration file **Matrikon.Datalogger.Collector.Service.exe.config**.
<Install Location>:\Program Files (x86)\Matrikon\Data Concentrator
3. Copy the contents of the installation media to a local drive.
4. Navigate to the copy of the installation media on the local drive.
Right-click **Setup.exe** and select **Run as administrator**.
5. Select **Data Concentrator Upgrade** and click **Next**.
6. Click **Install**.

If the Data Concentrator service is running, a command window will launch to shut it down. Allow the process to complete.



```

C:\Windows\system32\cmd.exe

SERVICE_NAME: Matrikon Data Concentrator
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 3   STOP_PENDING
                                <STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN>
        WIN32_EXIT_CODE       : 0   (0x0)
        SERVICE_EXIT_CODE   : 0   (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0
Allowing 30 seconds for process to clean up, please wait _

```

7. The Data Concentrator installation wizard launches.
8. Click **Next**.
9. Enter the folder to which Data Concentrator was installed, and click **Next**.
10. Click **Next** to continue the upgrade.

4.2 Upgrading the Forwarding Agent

4.2.1 Prerequisites

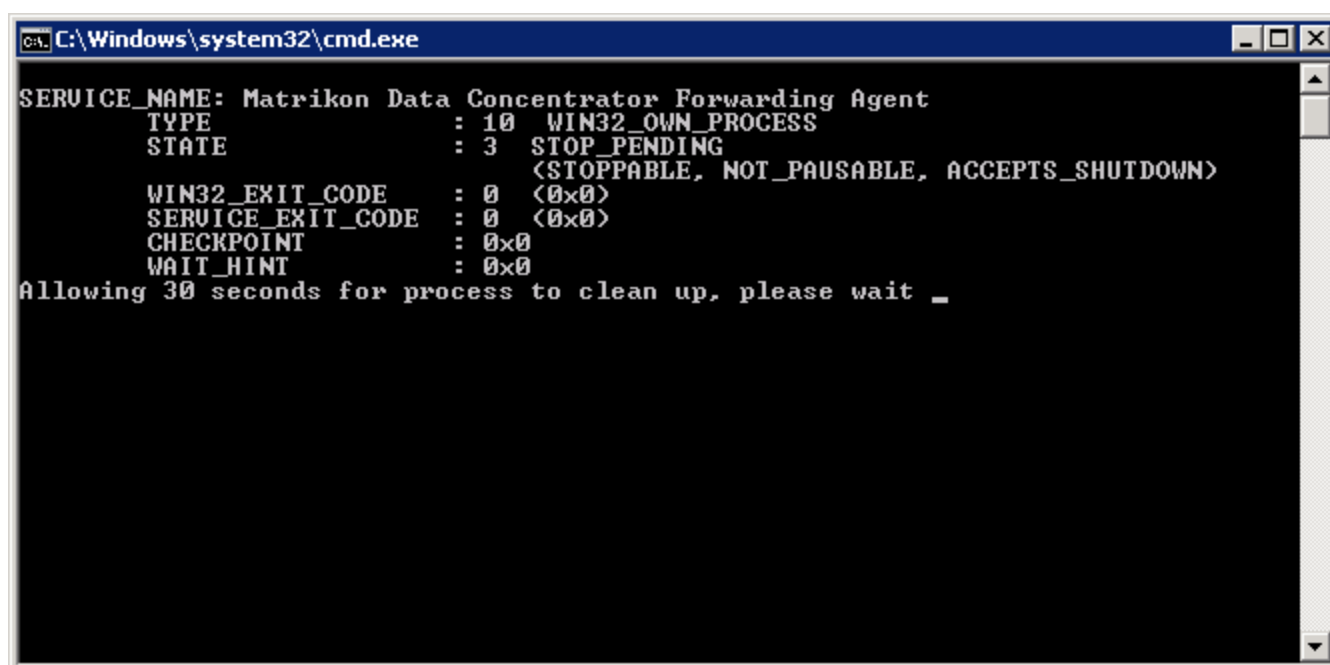
Following are requirements to upgrade the Forwarding Agent:

1. Forwarding Agent should be correctly installed and configured.
2. User with administrative privileges to perform the install.
3. Installation media.

4.2.2 Upgrade Procedure

1. Login as an administrative user in the computer with Forwarding Agent .
2. Navigate to the following folder and take a backup of Forwarding Agent configuration file **Matrikon.Datalogger.ForwardAgent.Service.exe.config**.
 <Install Location>:\Program Files (x86)\Matrikon\Data Concentrator Forwarding Agent
3. Copy the contents of the installation media to a local drive.
4. Navigate to the copy of the installation media on the local drive.
 Right-click **Setup.exe** and select **Run as administrator**.
5. Select **Forwarding Agent Upgrade**, and click **Next**.
6. Click **Install**.

If the Forwarding Agent service is running, a command window will launch to shut it down. Allow the process to complete.



```

C:\Windows\system32\cmd.exe

SERVICE_NAME: Matrikon Data Concentrator Forwarding Agent
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 3   STOP_PENDING
                                (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
        WIN32_EXIT_CODE       : 0   (0x0)
        SERVICE_EXIT_CODE   : 0   (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0
Allowing 30 seconds for process to clean up, please wait _

```

7. The Forwarding Agent installation wizard launches.
8. Click **Next**.
9. Enter the folder to which Forwarding Agent 140.1 was installed, and click **Next**.
10. Click **Next** to continue the upgrade.

4.3 Renaming the Data Concentrator Server

Do all of the following to update the Data Concentrator if the name of the server is changed and, if required, to change the services user.

Follow the below steps to update the Data Concentrator, if the name of the server is updated. If required, to change the services user.

1. Navigate to the following folder in the installation media:
 <Installation Media>:\Installer\NodeRenameDataConcentrator\
2. Copy the **DC_Rename.ps1** file to a location in the Data Concentrator server.
3. Right-click the copied **DC_Rename.ps1** file and select **Run as Administrator**.
4. When prompted, type the following:
 - a. The old name of the Data Concentrator server without the domain name.
 - b. The new name of the Data Concentrator server without the domain name.
 - c. The old name of the Data Concentrator server with the domain name.
 - d. The new name of the Data Concentrator server with the domain name.
 - e. The name of the new services user. (Leave this empty to skip the services user change).
 - f. The password of the new services user.
5. If the services user has change, open SQL Management Studio and ensure the new services user has the following permissions to the Data Concentrator database:
 - a. db_darareader
 - b. db_datawriter

DATA CONCENTRATOR CONFIGURATION

5.1 Data Concentrator Configuration File

Most of the Data Concentrator configuration file is stored in the below location:

```
<Install Location>:\Program Files (x86)\Matrikon\Data Concentrator\Matrikon.Datalogger.Collector.exe.config
```

The .config file is an XML document, divided into different sections.

NOTE

For any configuration changes to take effect, the Matrikon Data Concentrator service should be restarted.

5.1.1 Global Configuration

Under the <applicationSettings> section of the configuration file are a number of configuration settings that are global to the Data Concentrator. In a number of cases, these settings may be overridden when specified on a Datalogger by Datalogger basis.

NOTE

Certain reserved XML characters (such as angle brackets) need to be suitably escaped.

For example, '<' needs to be represented as '<'.

Parameter	Description	Example
ArchiveFileLocation	The absolute path on the file system which will be the 'root' of the archive file structure. Under this directory will be further subdirectories for the Sensor, Alarm and Log files, and files will be moved as appropriate after successful processing.	C:\Datalogger\Archive
BadFileLocation	The absolute path on the file system which will be the 'root' of the 'bad' file structure.	C:\Datalogger\Bad

Parameter	Description	Example
	Under this directory will be further subdirectories for the Sensor, Alarm and Log files. Files will be moved as appropriate after failed processing.	
SensorFilePattern	A regular expression pattern that if matched, identifies a file as a Sensor file.	_....._..._.*?_Realtime
AlarmFilePattern	A regular expression pattern that if matched, identifies a file as an Alarm file.	_....._..._.*?_Alarm
LogFilePattern	A regular expression pattern that if matched, identifies a file as a Log file.	_....._..._.*?_Log
SensorFileQualifier	A regular expression pattern that identifies the qualifier (i.e., site/mine/equipment) from a file name. This qualifier (if any) is used to resolve full historian tag names from tag aliases used inside Datalogger sensor data files.	(?&lt;=....._....._..._). *? (?=_Realtime)
AlarmFileQualifier	A regular expression pattern that identifies the qualifier (i.e., site/mine/equipment) from a file name. This qualifier (if any) is used to prefix Alarm messages before they are written to Alarm Manager.	(?&lt;=....._....._..._). *? (?=_Alarm)
LogFileQualifier	A regular expression pattern that identifies the qualifier (i.e., site/mine/equipment) from a file name. This qualifier (if any) is	(?&lt;=....._....._..._). *? (?=_Log)

Parameter	Description	Example
	used to prefix Log messages before they are written to Alarm Manager.	
RelativeSensorArchivePath	The directory under the root archive location where Datalogger sensor data files are moved after successful processing.	Sensor
RelativeAlarmArchivePath	The directory under the root archive location where Datalogger alarm data files are moved after successful processing.	AE
RelativeLogArchivePath	The directory under the root archive location where Datalogger log data files are moved after successful processing.	Log
RelativeOtherArchivePath	The directory under the root archive location where 'Other' (i.e., non-Sensor/Alarm/Log) files are moved after successful processing.	Other
UOMConversionFile	Absolute path to the file containing unit conversion definitions. Refer Unit Conversions on how it is configured.	C:\Program Files (x86)\Matrikon\Data Concentrator\UOMConversions.csv
AMCollectorHost	The hostname for the server on which the Alarm Manager TCP/IP Collector Channel is installed.	localhost
AMCollectorPort	The port on which the Alarm Manager TCP/IP Collector Channel is listening.	22070

Parameter	Description	Example
StateSinkAsync	A flag indicating whether statistics should be written asynchronously or not. This will improve performance at risk of lost statistics when the service stops unexpectedly.	False
TagConfigRefreshMinutes	The time in minutes after which the tag configuration information is refreshed from the tag configuration database.	60
ArchiveLogMessagePattern	A regular expression pattern that if matched, identifies Datalogger Log messages important enough for archiving in Alarm Manager.	\[Diag\]
UTCTimestamps	A flag indicating if the system should treat incoming data as being UTC timestamped. If this is specified as False, the system will treat the incoming data as being in the local system time zone.	True
LockRetryMS	The time (in milliseconds) that the Data Concentrator waits to retry opening a file if the file is locked (for example, by another application).	10
MaxLockRetries	The number of times the Data Concentrator will retry opening a file if the file is locked (for example, by another application).	1000

Parameter	Description	Example
	The maximum amount of time (in ms) the Data Concentrator will try to open a locked file before giving up is: <code>LockRetryMS * MaxLockRetries</code>	
FileTimestampPattern	A regular expression identifying the timestamp portion in a filename.	<code>^....._....._...</code>
FileTimestampFormat	The format of the file timestamp, using .NET date format tokens. Please see Appendix A for a list of .NET date format tokens.	<code>yyyyMMdd_HHmss_fff</code>
TimeSyncThresholdMS	The threshold (in milliseconds) that the Data Concentrator will use to determine if a file has arrived out-of-order, or if the file was produced by a Data Logger that has lost time synchronization. For example, If this value is set to 5000, if a new file arrives that is older than the last processed file (for that Data Logger and file type) by more than 5 seconds, the file is considered 'bad', and will be moved to	5000

Parameter	Description	Example
	the 'BAD' archive location.	
EnforceTimeSync	A flag indicating whether or not the system should enforce time synchronization on the files that it receives. If set to False, the Data Concentrator will process incoming files regardless of the relative timestamp differences between files.	True
FutureTimeSyncThresholdMS	The threshold (in milliseconds) that the Data Concentrator uses to determine if a file has arrived too far in the future. For example, If this value is set to 5000, if a file arrives with a timestamp more than 5 seconds in the future (compared to 'now') at the time of processing, the file will be flagged as 'bad'.	5000
EnforceFutureTimeSync	A flag indicating whether or not the system should enforce future time synchronization on the files that it	False

Parameter	Description	Example
	receives.	
LogTypePattern	A regular expression identifying the type of log file from a log filename. This is necessary because as of version 2.1, the Data Concentrator tracks the last processed files for each <i>type</i> of log file for each Data Logger.	<pre>(?<=....._....._..._.*?_).*? (?=.log)</pre> Example: The above pattern applied against the following file: <i>20160206_001147_645_M2X_DLDTA.log</i> Will isolate 'DLDTA' as the log type.
RecoverableErrorRetryMS	The time (in milliseconds) that the Data Concentrator waits before it retries the processing of a failed file.	5000
LimitThreads	Specify False if the Data Concentrator should use a thread for processing each equipment. Specify 'True' to use a capped thread pool for processing.	False
MaxThreads	If LimitThreads is enabled, specifies the maximum number of threads that can run simultaneously.	32
EnableDataAgeCounters	Specifies whether or not to write the performance counters.	False
DataAgeEvaluationIntervalSeconds	This is the interval (in seconds) that the system will evaluate the data age counters, defaulted to 0. 0 means disabled.	0
SDHFileCacheMaximumAgeSeconds	This is the interval (in seconds) that the system will evaluate the system data health counters,	0

Parameter	Description	Example
	defaulted to 0. 0 means disabled.	
SDHFileCacheMaximumAgeSeconds	This is the size of the cache (in seconds) that the system will maintain for calculating the aggregated system data health counters, defaulted to 300.	300
InstanceName	This is the name of this installed data concentrator. Used in the counter instance names for the aggregated system data health counters. Each install within one deployment must be unique.	DC1
AggregateCounterEvaluationIntervalSeconds	This is the interval (in seconds) that the system will evaluate the aggregated system data health counters, defaulted to 0. 0 means disabled.	0
MaxRecoverableErrorRetries	This is the maximum number of times the system will retry a file that fails processing before skipping the file due to a “Recoverable” error, defaulted to 0. 0 means disabled (retry indefinitely, backwards compatible operation). This is not recommended for use as any server outage would result in files being skipped.	0
PHDRecoverableErrorCodes	These are the error code that the PHDAPI output treats as “Recoverable”.	

Parameter	Description	Example
MaxSuspectRecoverableErrorRetries	This is the maximum number of times the system will retry a file that fails processing before skipping the file due to a "SuspectRecoverable" error, defaulted to 0. 0 means disabled (retry indefinitely, backwards compatible operation).	5
SQLRecoverableErrorCodes	These are the error code that the SQL output treats as "Recoverable".	
SQLConnectionPoolSize	This is the size of the connection pool used by the SQL output.	8
SQLConnectionLeaseTimeoutMs	This is the connection lease timeout (in milliseconds) used by the SQL output.	30000
HistorianWriterFactoryTokenSeperator	This is the token separator for the 'Config' field in the Historian table of the DC database.	
HistorianWriterFactoryValueSeperator	This is the token name/value separator for the 'Config' field in the Historian table of the DC database.	?
UseSQLCETracking	Enable this to use SQL CE to store the tracking database. The default is SQL (the DataConcentrator database). Changing this setting would only be done in an effort to affect performance.	False
EnableStatsSink	Enable this to record stats for file processing.	False

Parameter	Description	Example
MaxNumberOfFilesToProcessPerBatch	The maximum number of incoming files per asset the system can process at a time. Adjusting this number can help to improve performance during file backlogs.	10
MonitoredCounterTypes	Enables the DC counters on the configured types. Comment out an asset type to disable the counters for that type.	SensorAELogOther
MonitoredAggregateCounterTypes	Enables the aggregate DC counters. Comment out an aggregate type to disable that aggregate counter.	AverageSumMaximumMinimum
TemplateAssetNameReplacementString	A replacement string in the for DataLogger templates. Is replaced by the asset name (or qualifier) for each instance the template generates.	{AssetName}
DefaultTemplateDataloggerID	The ID of the default datalogger template to use for the APM autosync.	UASTemplateDatalogger
WaitForNextFilesMilliseconds	Configurable number of seconds that the Data Concentrator waits before checking for new files.	500
TemplateInstanceNameReplacementString	The replacement string for instance in the DataLogger templates. Is replaced by the data concentrator instance name for each instance the template generates.	{InstanceName}
IncludeSubdirectories	The default setting for whether or not to	False

Parameter	Description	Example
	include subdirectories when looking for files to consume. NOTE Setting this value to True can become resource intensive on systems with many subdirectories.	

5.1.2 Connection Strings

The SQL Server connection string, which allows connection to the SQL Server database instance, is located in the **<connectionStrings>** section as given below:

```
<connectionStrings>

<add name="DataConcentratorConnectionString" connectionString="Initial
Catalog=DataConcentrator;DataSource=(local)\SQLEXPRESS;User=sa;Password= pword"
providerName="System.Data.SqlClient"/>

<add name="Entities"
connectionString="metadata=res://*/Tracking.csdl|res://*/Tracking.ssdl|res://*/Tracki
ng.msl;provider=System.Data.SqlClient;provider connection string="
[DataConcentratorConnectionString]" providerName="System.Data.EntityClient" />

<add
name="DCUEntities"connectionString="metadata=res://*/DCU.csdl|res://*/DCU.ssdl|res://
*/DCU.msl;provider=System.Data.SqlClient;provider connection string="
[DataConcentratorConnectionString]" providerName="System.Data.EntityClient" />

</connectionStrings>
```

For more information on formatting a connection string, refer [Connection String Formats](#).

5.1.3 Global Input Configuration

As of Data Concentrator 2.4, each input handler can be configured globally. At the time of this writing, the default sensor handler (i.e., Matrikon Datalogger Input Handler) and Multi Sensor Input Handler expose configuration parameters:

```
<sensorInputHandlers>

<defaultSensorInputHandler>

<param key="dateFormat" value="dd-MM-yyyy HH:mm:ss.fff"/>

</defaultSensorInputHandler>

</sensorInputHandlers>

<alarmInputHandlers>

<defaultAlarmInputHandler />
```

```
</alarmInputHandlers>

<logInputHandlers>

<defaultLogInputHandler />

</logInputHandlers>
```

Parameter	Description	Example
sensorInputHandlers/defaultSensorInputHandler/param [@key=dateFormat]	The date format (in .NET syntax) expected for dates in Text/CSV datalogger files. See Appendix A for a list of .NET date/time identifiers.	dd-MM-yyyy HH:mm:ss.fff

The Multi Sensor Input Handler allows for the parsing of concatenated zip files (for example, you can configure the Data Concentrator to send these after a network outage), as well as the functionality of the default sensor handler. Only one handler should be used at a time for a given asset.

5.1.4 Datalogger Connection Configuration

It is expected that each Datalogger is configured to FTP files to a different set of directories. Following is an example for directory configuration:

- Datalogger 1
 - Sensor
 - AE
 - Log
- Datalogger 2
 - Sensor
 - AE
 - Log

When configured this way, each Datalogger will receive equal data processing resources regardless of the file backlog in a given directory.

NOTE

When using the APM autosync, a templated datalogger configuration is used. In this case, you do not need to create a datalogger configuration section.

The **<dataloggerConfig>** section are the specific configuration settings for each datalogger, and the directories that are monitored. A Datalogger definition (for a fictional 'M2X' equipment) resembles the following:

```
<datalogger ID="M2X">

<amCollectorHost>localhost</amCollectorHost>

<amCollectorPort>22070</amCollectorPort>

<archiveLogMessagePattern>\[Diag\]</archiveLogMessagePattern>

<archiveFileLocation>C:\Datalogger\Archive</archiveFileLocation>
```

```

<badFileLocation>C:\Datalogger\Bad</badFileLocation>
<utcTimestamps>False</utcTimestamps>
<enforceTimeSync>True</enforceTimeSync>
<timeSyncThresholdMS>5000</timeSyncThresholdMS>
<enforceFutureTimeSync>False</enforceFutureTimeSync>
<futureTimeSyncThresholdMS>3600000</futureTimeSyncThresholdMS>
<timeZone>Eastern Standard Time</timeZone>
<mode>realtime</mode>
<numberOfRealTimeFiles>0</numberOfRealTimeFiles>
<alarmFileConfig handlerTypeName="defaultAlarmInputHandler">
  <directory>C:\M2X\Alarm</directory>
  <priority>1</priority>
  <filter>*Alarm.*</filter>
  <pattern>....._....._..._.*?_Alarm</pattern>
  <qualifier>(?!&lt;=....._....._..._).*(?=_Alarm)</qualifier>
  <timestampPattern>^....._....._...</timestampPattern>
  <timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
</alarmFileConfig>
<sensorFileConfig handlerTypeName="defaultSensorInputHandler">
  <directory>C:\M2X\Sensor</directory>
  <priority>2</priority>
  <filter>*Realtime.*</filter>
  <pattern>....._....._..._.*?_Realtime</pattern>
  <qualifier>(?!&lt;=....._....._..._).*(?=_Realtime)</qualifier>
  <timestampPattern>^....._....._...</timestampPattern>
  <timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
  <typePattern>Realtime</typePattern>
  <extendedConfiguration>
    <param key="dateFormat" value="dd-MM-yyyy HH:mm:ss.fff"/>
  </extendedConfiguration>
</sensorFileConfig>
<logFileConfig handlerTypeName="defaultLogInputHandler">

```

```
<directory>C:\M2X\Log</directory>

<priority>3</priority>

<filter>*. *</filter>

<pattern>....._....._..._.*?_Log</pattern>

<qualifier>(?!&lt;=....._....._..._). *?(?=_Log)</qualifier>

<timestampPattern>^....._....._...</timestampPattern>

<timestampFormat>yyyyMMdd_HHmss_fff</timestampFormat>

<typePattern>(?!&lt;=....._....._..._.*?_). *?(?=.log)</typePattern>

</logFileConfig>

</datalogger>
```

A **datalogger** entry will exist for each Datalogger under the **<dataloggerConfig>** section. Each datalogger entry has the following configuration options:

Parameter	Description	Example
ID	A unique identifier for the datalogger.	M2X
amCollectorHost	(Optional) The hostname for the server on which the Alarm Manager TCP/IP Collector Channel is installed. If specified, overrides the AMCollectorHost specified in the global configuration section.	localhost
amCollectorPort	(Optional) The port on which the Alarm Manager TCP/IP Collector Channel is listening. If specified, overrides the AMCollectorPort specified in the global configuration section.	22070
archiveLogMessagePattern	(Optional) A regular expression pattern that if matched, identifies Datalogger Log messages important enough for archiving in Alarm Manager.	\[Diag\]

Parameter	Description	Example
	If specified, overrides the ArchiveLogMessagePattern specified in the global configuration section.	
archiveFileLocation	(Optional) The absolute path on the file system which will be the 'root' of the archive file structure. Under this directory will be further subdirectories for the Sensor, Alarm and Log files, and files will be moved as appropriate after successful processing. If specified, overrides the ArchiveFileLocation specified in the global configuration section.	C:\Datalogger\Archive
badFileLocation	(Optional) The absolute path on the file system which will be the 'root' of the 'bad' file structure. Under this directory will be further subdirectories for the Sensor, Alarm and Log files. Files will be moved as appropriate after failed processing. If specified, overrides the BadFileLocation specified in the global configuration section.	C:\Datalogger\Bad
utcTimestamps	A flag indicating if the system should treat incoming data as being UTC timestamped. If this is specified as False, the system will treat the incoming data as being in the local system time zone. If specified, overrides the UTCTimestamps specified in the global configuration section.	True
enforceTimeSync	(Optional)	True

Parameter	Description	Example
	A flag indicating whether or not the system should enforce time synchronization on the files that it receives. If set to False, the Data Concentrator will process incoming files regardless of the relative timestamp differences between files. If specified, overrides the EnforceTimeSync specified in the global configuration section.	
timeSyncThresholdMS	(Optional) The threshold (in milliseconds) that the Data Concentrator will use to determine if a file has arrived out-of-order, or if the file was produced by a Data Logger that has lost time synchronization. For example, if this value is set to 5000, if a new file arrives that is older than the last processed file (for that Data Logger and file type) by more than 5 seconds, the file is considered 'bad', and will be moved to the 'BAD' archive location. If specified, overrides TimeSyncThresholdMS in the global configuration section.	5000
enforceFutureTimeSync	(Optional) A flag indicating whether or not the system should enforce future time synchronization on the files that it receives. If specified, overrides the EnforceFutureTimeSync in the global configuration section.	False
futureTimeSyncThresholdMS	(Optional)	5000

Parameter	Description	Example
	The threshold (in milliseconds) that the Data Concentrator uses to determine if a file has arrived too far in the future. For example, if this value is set to 5000, if a file arrives with a timestamp more than 5 seconds in the future (compared to 'now') at the time of processing, the file will be flagged as 'bad'. If specified, overrides FutureTimeSyncThreshold MS in the global configuration section.	
timeZone	(Optional) The Id of the time zone for this datalogger. This instructs the Data Concentrator to interpret incoming files as containing data timestamped in 'local' times in the specified time zone. Refer Supported Time Zones for a list of available time zones. NOTE If the timeZone is specified and utcTimestamps is set to true for this datalogger, the timeZone will be ignored.	Eastern Standard Time
alarmFileConfig/directory	The absolute directory to monitor for this Datalogger's alarm files.	C:\M2X\Alarm
alarmFileConfig/priority	The priority of Alarm files relative to other types of files. The lower the number, the higher the priority. It is recommended that this value remains '1', unless a Datalogger has	1

Parameter	Description	Example
	special requirements.	
alarmFileConfig/filter	The filter affecting the types of files monitored in the directory. If you know the extension of Alarm files produced by this Datalogger, please specify it here.	*.pgd
alarmFileConfig/pattern	(Optional) A regular expression pattern that if matched, identifies a file as an Alarm file. If specified, overrides the AlarmFilePattern specified in the global configuration setting.	_....._..._.*?_Alarm
alarmFileConfig/qualifier	(Optional) A regular expression pattern that identifies the qualifier (i.e., site/mine/equipment) from a file name. This qualifier (if any) is used to prefix Alarm messages before they are written to Alarm Manager. If specified, overrides the AlarmFileQualifier specified in the global configuration setting.	(?<=....._....._..._). *? (?=_Alarm)
alarmFileConfig/timestampPattern	(Optional) A regular expression identifying the timestamp portion in an alarm file filename. If specified, overrides the FileTimestampPattern in the global configuration section, for alarm files.	^....._....._...
alarmFileConfig/timestampFormat	(Optional) The format of the file timestamp in an alarm file filename, using .NET date format tokens. Please see Appendix A for a list of	yyyyMMdd_HHmss_fff

Parameter	Description	Example
	.NET date format tokens. If specified, overrides the FileTimestampFormat in the global configuration section, for alarm files.	
sensorFileConfig/directory	The absolute directory to monitor for this Datalogger's alarm files.	C:\M2X\Alarm
sensorFileConfig/priority	The priority of Sensor files relative to other types of files. The lower the number, the higher the priority. It is recommended that this value remains '2', unless a Datalogger has special requirements.	2
sensorFileConfig/filter	The filter affecting the types of files monitored in the directory. If you know the extension of Sensor files produced by this Datalogger, please specify it here.	*.bin
sensorFileConfig/pattern	(Optional) A regular expression pattern that if matched, identifies a file as a Sensor file. If specified, overrides the SensorFilePattern specified in the global configuration setting.	_....._..._.*?_ Realtime
sensorFileConfig/qualifier	(Optional) A regular expression pattern that identifies the qualifier (i.e., site/mine/equipment) from a file name. This qualifier (if any) is used to resolve full historian tag names from tag aliases used inside Datalogger sensor data files. If specified, overrides the SensorFileQualifier specified in the global	(?<=....._....._..._).*? (?=_Realtime)

Parameter	Description	Example
	configuration setting.	
sensorFileConfig/timestampPattern	(Optional) A regular expression identifying the timestamp portion in a sensor file filename. If specified, overrides the FileTimestampPattern in the global configuration section, for sensor files.	^....._....._...
sensorFileConfig/timestampFormat	(Optional) The format of the file timestamp in a sensor file filename, using .NET date format tokens. Please see Appendix A for a list of .NET date format tokens. If specified, overrides the FileTimestampFormat in the global configuration section, for sensor files.	yyyyMMdd_HHmss_fff
sensorFileConfig/extendedConfiguration/param[@key=dateFormat]	(Optional - Applicable only for the <i>defaultSensorInputHandler</i>). The date format (in .NET syntax) expected for dates in Text/CSV datalogger files. Refer .NET Date Format Tokens for a list of .NET date/time identifiers. If specified, overrides the dateFormat specified in the global configuration for the defaultSensorInputHandler.	dd-MM-yyyy HH:mm:ss.fff
logFileConfig/directory	The absolute directory to monitor for this Datalogger's alarm files.	C:\M2X\Alarm
logFileConfig/priority	The priority of Log files relative to other types of files. The lower the number, the higher the priority. It is recommended	3

Parameter	Description	Example
	that this value remains '3', unless a Datalogger has special requirements.	
logFileConfig/filter	The filter affecting the types of files monitored in the directory. If you know the extension of Log files produced by this Datalogger, please specify it here.	*.log
logFileConfig/pattern	(Optional) A regular expression pattern that if matched, identifies a file as a Log file. If specified, overrides the LogFilePattern specified in the global configuration setting.	_....._..._.*?_Log
logFileConfig/qualifier	(Optional) A regular expression pattern that identifies the qualifier (i.e., site/mine/equipment) from a file name. This qualifier (if any) is used to prefix Log messages before they are written to Alarm Manager. If specified, overrides the LogFileQualifier specified in the global configuration setting.	(?<=....._....._..._).*? (?=_Log)
logFileConfig/timestampPattern	(Optional) A regular expression identifying the timestamp portion in a log file filename. If specified, overrides the FileTimestampPattern in the global configuration section, for log files.	^....._....._...
logFileConfig/timestampFormat	(Optional) The format of the file timestamp in a log file filename, using .NET date format tokens.	yyyyMMdd_HHmss_fff

Parameter	Description	Example
	Please see Appendix A for a list of .NET date format tokens. If specified, overrides the FileTimestampFormat in the global configuration section, for log files.	
logFileConfig/typePattern	A regular expression identifying the type of log file from a log filename. This is necessary because as of version 2.1, the Data Concentrator tracks the last processed files for each <i>type</i> of log file for each Data Logger. If specified, overrides LogTypePattern in the global configuration section.	<pre>(?&lt; =....._....._..._.*?_).*? (?=.log)</pre> Example: The above pattern applied against the following file: <code>20160206_001147_645_M2X_DLDTA.log</code> will isolate 'DLDTA' as the log type.
mode	This settings determines the mode that the system runs in. In realtime mode, the system will only process the X newest files it finds in the folder specified (Once it processes a file, it will not process anything older). In historical mode it does the opposite and only process all files EXCEPT the X newest files it finds in the folder specified. These two options are meant to be used at the same time to allow offloading of large numbers of backlogged files to be sent to a separate historian / Asset Performance Management data store for performance reasons. If it is imperative that the most recent data is processed ASAP or if the system only require the most recent data (and not all data	realtime or historical

Parameter	Description	Example
	needs to be processed), then use realtime mode only. All files are still processed from oldest to newest within the specified mode. X = numberOfRealTimeFiles	
numberOfRealTimeFiles	This is the number of realtime files that realtime and historical mode will base their processing on. 0 = Handle all file as if in realtime mode (this is the standard operation of the system).	Any integer greater than or equal to 0
templateOnly	Specifies the datalogger configuration is a template.	True or False
enableArchive	Enables or disables forwarding processed files to the <archiveFileLocation> directory. True = default	True or False
monitorSubdirectories	Enables the Data Concentrator to <i>only</i> monitor the child directories of the <directory> directory. Result in a thread for each child directory and monitors for newly created, renamed, deleted child directories. False = default	True or False
IncludeSubdirectories	The default setting for whether or not to include subdirectories when looking for files to consume. NOTE Setting this value to True can become resource intensive on systems with many subdirectories.	False

5.1.5 Tag Configuration Source

The <tagProviderConfig> section provides settings for the tag configuration component. As of Data Concentrator v2.2, the default tag configuration provider supports the following:

```
<tagProviderConfig>

<tagConfigRefreshMinutes>60</tagConfigRefreshMinutes>

<retrySeconds>600</retrySeconds>

<uasHistorianTypeName>UAS</uasHistorianTypeName>

</tagProviderConfig>
```

Parameter	Description	Example
tagConfigRefreshMinutes	The time in minutes after which the tag configuration information is refreshed from the tag configuration database.	60
retrySeconds	The time in seconds use to wait before retrying a connection that has failed.	600
uasHistorianTypeName	The historian type used by the autosync process. Currently supported: APM.	APM

5.1.6 File Ordering

For each input (i.e. directory) configured for a datalogger, the files are internally ordered **chronologically** using the timestamps extracted from the filenames using the **timestamp pattern** and **timestamp format** configured for that input location.

5.1.7 Priority Files

It is possible to support multiple levels of priorities within a type of file. For example, consider a case where a datalogger writes files to the following structure:

Incoming Folder	File Type	Desired Processing Priority
Alarm	Datalogger alarm files	2
Sensor	Datalogger sensor files	3
Priority	High-priority Datalogger sensor files	1
Log	Datalogger log files	4

This example can be achieved by configuring an additional input configuration for **Priority** files with a higher priority:

```
<alarmFileConfig handlerTypeName="defaultAlarmInputHandler">

<directory>C:\Input\Alarm</directory>

<priority>2</priority>

<filter>*.*</filter>

<pattern>....._....._..._.*?_Alarm</pattern>

<qualifier>(?!<=....._....._..._).*(?=_Alarm)</qualifier>

<timestampPattern>^....._....._...</timestampPattern>
```

```

<timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
</alarmFileConfig>

<sensorFileConfig handlerTypeName="defaultSensorInputHandler">
<directory>C:\Input\Sensor</directory>
<priority>3</priority>
<filter>*Realtime.*</filter>
<pattern>....._....._..._.*?_Realtime</pattern>
<qualifier>(?!<=....._....._..._).*(?=_Realtime)</qualifier>
<timestampPattern>^....._....._...</timestampPattern>
<timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
<typePattern>Realtime</typePattern>
<extendedConfiguration>
<param key="dateFormat" value="dd-MM-yyyy HH:mm:ss.fff"/>
</extendedConfiguration>
</sensorFileConfig>

<sensorFileConfig handlerTypeName="defaultSensorInputHandler">
<directory>C:\Input\Priority</directory>
<priority>1</priority>
<filter>*Priority.*</filter>
<pattern>....._....._..._.*?_Priority</pattern>
<qualifier>(?!<=....._....._..._).*(?=_Priority)</qualifier>
<timestampPattern>^....._....._...</timestampPattern>
<timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
<typePattern>Priority</typePattern>
<extendedConfiguration>
<param key="dateFormat" value="dd-MM-yyyy HH:mm:ss.fff"/>
</extendedConfiguration>
</sensorFileConfig>

<logFileConfig handlerTypeName="defaultLogInputHandler">
<directory>C:\Input\Priority</directory>
<priority>4</priority>
<filter>*. *</filter>

```

```
<pattern>....._....._..._.*?_Log</pattern>

<qualifier>(?!<=....._....._..._).*(?=_Log)</qualifier>

<timestampPattern>^....._....._...</timestampPattern>

<timestampFormat>yyyyMMdd_HH:mm:ss_fff</timestampFormat>

<typePattern>(?!<=....._....._..._.*?_).*(?=\.log)</typePattern>

</logfileConfig>
```

Please note the following if configuring additional inputs for a datalogger:

- The file names should have some distinguishing features (i.e., normal sensor files contain **Realtime** in the filename while priority sensor files contain **Priority** even though the content structure is the same).
- More than one input handler can monitor a single directory, but be sure to make the **filter** wildcard as specific as possible to avoid conflicts between input handlers.
The above example shows normal sensor files processed with a ***Realtime.*** filter while priority files are processed with a ***Priority.*** filter.
- When multiple handlers are specified for one type of file, each of those input handlers **must** specify a **typePattern** that resolves to a unique value. The reason for this is because the Data Concentrator tracks the **last processed file** separately for each file class (i.e., sensor / alarm / log) and file subclass (e.g., Priority).

Consider a typical configuration with just 3 types of files (i.e., ALARM, SENSOR and LOG). Internally, the Data Concentrator will track the files for 'M2X' as such:

Datalogger ID	File Type	Last Processed File
M2X	ALARM	...
M2X	SENSOR	...
M2X	LOG	...

Now consider a configuration that includes the new type of **Priority** sensor files. Internally, the Priority files need to be tracked independently of the regular sensor files:

Datalogger ID	File Type	Last Processed File
M2X	ALARM	...
M2X	SENSOR / REALTIME	...
M2X	SENSOR / PRIORITY	...
M2X	LOG	...

The **REALTIME** and **PRIORITY** file subtypes come from the **typePattern** input configuration parameter.

5.1.8 VIMS Support (Experimental)

Data Concentrator v2.4 supports the processing of VIMS files. Add **other** to the input entry (shown in bold below) to the datalogger configuration to configure VIMS file processing.

```

    <datalogger ID="M2X">
    <amCollectorHost>localhost</amCollectorHost>
    <amCollectorPort>22070</amCollectorPort>
    <archiveLogMessagePattern>\[Diag\]</archiveLogMessagePattern>
    <archiveFileLocation>C:\Datalogger\Archive</archiveFileLocation>
    <badFileLocation>C:\Datalogger\Bad</badFileLocation>
    <utcTimestamps>False</utcTimestamps>
    <enforceTimeSync>True</enforceTimeSync>
    <timeSyncThresholdMS>5000</timeSyncThresholdMS>
    <enforceFutureTimeSync>False</enforceFutureTimeSync>
    <futureTimeSyncThresholdMS>3600000</futureTimeSyncThresholdMS>
    <timeZone>Eastern Standard Time</timeZone>
    <mode>realtime</mode>
    <numberOfRealTimeFiles>0</numberOfRealTimeFiles>
    <alarmFileConfig handlerTypeName="defaultAlarmInputHandler">
    <directory>C:\M2X\Alarm</directory>
    <priority>1</priority>
    <filter>*Alarm.*</filter>
    <pattern>....._....._..._.*?_Alarm</pattern>
    <qualifier>(?!<=....._....._..._).*(?=_Alarm)</qualifier>
    <timestampPattern>^....._....._...</timestampPattern>
    <timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
    </alarmFileConfig>
    <sensorFileConfig handlerTypeName="defaultSensorInputHandler">
    <directory>C:\M2X\Sensor</directory>
    <priority>2</priority>
    <filter>*Realtime.*</filter>
    <pattern>....._....._..._.*?_Realtime</pattern>
    <qualifier>(?!<=....._....._..._).*(?=_Realtime)</qualifier>
    <timestampPattern>^....._....._...</timestampPattern>
    <timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
    <typePattern>Realtime</typePattern>

```

```
<extendedConfiguration>
<param key="dateFormat" value="dd-MM-yyyy HH:mm:ss.fff"/>
</extendedConfiguration>
</sensorFileConfig>
<logfileConfig handlerTypeName="defaultLogInputHandler">
<directory>C:\M2X\Log</directory>
<priority>3</priority>
<filter>*. *</filter>
<pattern>....._....._...._*?_Log</pattern>
<qualifier>(?!<=....._....._...._)*?(?=_Log)</qualifier>
<timestampPattern>^....._....._....</timestampPattern>
<timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
<typePattern>(?!<=....._....._...._.*?_)atePattern>
</logfileConfig>
<otherFileConfig handlerTypeName="vimsInputHandler">
<directory>C:\M2X\VIMS</directory>
<priority>5</priority>
<filter>*. *</filter>
<pattern>....._....._...._*?_VIMS</pattern>
<qualifier>(?!<=....._....._...._)*?(?=_VIMS)</qualifier>
<timestampPattern>^....._....._....</timestampPattern>
<timestampFormat>yyyyMMdd_HH:mm:ss.fff</timestampFormat>
<typePattern>VIMS</typePattern>
<extendedConfiguration>
<param key="destinationPath" value="C:\VIMSOut"/>
</extendedConfiguration>
</otherFileConfig>
</datalogger>
```

The vimsInputHandler exposes the following configuration:

Parameter	Description	Example
/datalogger/otherFile Config/extendedConfiguration/ param[@key=destinationPath]	A path on the file system where processed VIMS files are written.	C:\VIMSOut

5.1.9 Logging

The Data Concentrator uses Apache log4net for logging, and supports the full range of log4net configuration parameters.

Documentation on the log4net configuration can be found in the below path:

<http://logging.apache.org/log4net/release/config-examples.html>

By default, the Data Concentrator uses the following basic configuration defined in the **<log4net>** section:

```
<log4net>

  <appender name="RollingFileAppender"
type="log4net.Appender.RollingFileAppender">

    <param name="File" value="C:/CollectorLog.txt" />
    <param name="AppendToFile" value="true" />
    <param name="RollingStyle" value="Size" />
    <param name="MaxSizeRollBackups" value="10" />
    <param name="MaximumFileSize" value="1024KB" />
    <param name="StaticLogFileName" value="true" />
    <layout type="log4net.Layout.PatternLayout">
      <param name="ConversionPattern" value="%d [%t] %-5p %c - %m%n" />
    </layout>
  </appender>

  <root>

    <level value="DEBUG" />
    <appender-ref ref="RollingFileAppender" />
  </root>
</log4net>
```

The **RollingFileAppender** will maintain a fixed set of log files (defined by the **MaxSizeRollBackups** and **MaximumFileSize** parameters), with a base file name defined by the **File** parameter. This configuration should be suitable for most situations.

The log **level** should also be defined as appropriate for your stage of the project. When first commissioning the system, leaving the level at **DEBUG** will make debugging problems easier, at the cost of a large amount of logging statements. In general production, a log level of **INFO** would be appropriate.

5.2 DCU Configuration

5.2.1 Historian Configuration

The Data Concentrator allows any number of OPC HDA servers to be used, in addition to PI UFL (Universal File and Stream Loader) output directories, PHD API, Asset Performance Management, or SQL. This configuration is stored in the **Historian** table in the SQL Server database.

The **Historian** table has 4 columns:

1. **ID:** An automatically generated ID for the historian definition.
2. **TypeID:** The ID of the historian type.

When the database schema is created, the following HistorianTypes are inserted and available for use:

- a. **OPC** (ID of 1)
- b. **UFL** (ID of 2)
- c. **PHDAPI** (ID of 3)
- d. **APM** (ID of 4)
- e. **SQL** (ID of 5)

3. **Name:** The name of the historian.

The value of this is unimportant, but should hold some significance to allow debugging of problems in the log files.

4. **Config:** The configuration/connection information for the historian.

This column supports either a name/value pair or xml format. XML is primarily intended for use when a Name/Value pair cannot be used (for example, when an = character must be included), and must be XML-escaped.

OPC

NOTE

To use OPC, the OPC Core Components Redistributable needs to be installed on the local computer. Run the following file from the installation media to install the OPC Core Components Redistributable:

<Install Media>: \Prerequisites\OPC Core Components Redistributable\setup.exe

OPC servers have the following configuration format:

Host=hostname;ProgID=progid;MaxTransactionRecords=n

Name/Value Pair Example:

```
"Host=MEM31;ProgID=Matrikon.OPC.IP21SetCim01.1;
MaxTransactionRecords=2147483647"
```

XML Example:

```
"<connectionxml><host>MEM31</host><progid>Matrikon.OPC.IP21SetCim01.1</progid><maxtransactionrecords>2147483647</maxtransactionrecords></connectionxml>"
```

NOTE

MaxTransactionRecords defines the maximum number of records for each historical write operation. If the total number of records in a file exceeds this value, the operation is broken into multiple writes. If this parameter is omitted, no maximum transaction size is enforced.

UFL

UFL entries have the following configuration format:

Directory=dirName;UTC=true|false;overwriteexistingfile=true|false

If **overwriteexistingfile** is **False** (default), a new UFL file with an incremented name is created, if one by the name already exists. If it is **True**, the file is overwritten.

Name/Value Pair Example:

```
"Directory=C:\Output1;UTC=true;overwriteexistingfile=true"
```

XML Example:

```
"<connectionxml><directory>
C:\Output1</directory><utc>true</utc></connectionxml>"
```

NOTE

Please ensure UFL is configured accordingly, if output has the following format:

Tag, Timestamp, Value, Quality

Where, the Timestamp is formatted as dd-MMM-yyyy hh:mm:ss.nnn

PHDAPI

NOTE

To use PHDAPI, the PHD Client needs to be installed on the local computer. Run the following file from the installation media to install the PHD Client:

<Install Media>:\Prerequisites\PHDClient321x64.msi

PHDAPI entries have the following configuration format:

Host=hostname;Port=port;Instances=instances;Batching=True

Where:

- **Host:** The name of the PHD server.
- **Port:** The port of the PHD server (default: 3100).
- **Instances:** The number of simultaneous connections to the PHD server (default: 10).
- **Batching:** Send the data as individual inserts (False), or as a batched insert (True). Only supported on PHD version 310 or higher.

Name/Value Pair Example:

```
"Host=MEM-PHD;Port=3100;Instances=10;Batching=True"
```

XML Example:

```
"<connectionxml><host>MEM-  
PHD</host><port>3100</port><instances>10</instances><batching>true</batching></c  
onnectionxml>"
```

After rows are entered in the **Historian** table for each historian to be used, tags can be configured as illustrated in the next section.

Asset Performance Management

NOTE

To ensure the Data Concentrator can access Asset Performance Management, the account the Data Concentrator is running should be added to the Reliability Engineers group.

For instructions on adding users to the Reliability Engineers group, refer the Post Installation Steps in the Asset Performance Management Installation Guide.

APM entries have the following configuration format:

HostUrl=hostUrl; UTC=true|false; stateRefreshSecond=ss; SyncFromUAS=true|false; removetagsnotdefinedinuas=true|false; enableUOMConversions= true|false;

Where:

- **HostUrl**: The URL of the APM Web Server, including port number.
- **UTC**: Whether or not the system should treat incoming data as being UTC timestamped (if false, local system time zone is used).
- **SyncFromUAS**: Enables the Data Concentrator to automatically synchronize its tag definitions from APM.
- **EnableUOMConversions**: Enables the Data Concentrator to perform unit conversions against data before sending it to APM.

Name/Value Pair Example:

```
"hosturl=http://localhost:80/MES/AM/; utc=True; stateRefreshSeconds=30;  
syncToUAS=True; removetagsnotdefinedinuas=True; enableUOMConversions=True;"
```

XML Example:

```
"<connectionxml><hosturl>http://webserver.com:80/MES/AM/</hosturl><utc>true</utc>  
<stateRefreshSeconds>30</stateRefreshSeconds><syncToUAS>true</syncToUAS><removeta  
gsnotdefinedinuas>true</removetagsnotdefinedinuas><enableUOMConversions>true</ena  
bleUOMConversions></connectionxml>"
```

Tag Synchronization

If APM autosync is enabled (syncToUAS=true in the above), the tag synchronization between APM and the DC is automatically handled, and manual configuration of the tags are not required.

If you enable APM autosync in a system with existing settings, those settings will be lost. For information on migrating your data, please contact Honeywell Customer Support.

1. Open Asset Performance Management.
2. Open the Application Configuration page (**Gear icon > System > System Configuration > Application**).
3. Locate the following setting:
Honeywell.UAS.Data.DC.SyncFromUAS.SettingName
4. Double-click the value box for the autosync setting, and enter **True**.
5. Click **Save**.
6. Follow the below steps:
 - a. On the Contents tree, select **System Console**.
 - b. In the Publish Saved Changes section, click **Publish**.

To synchronize Data Concentrator with Data Logger, follow the below steps:

1. Open Asset Performance Management.
2. Open the Application Configuration page (**Gear icon > System > System Configuration > Application**).
3. Locate the following setting and update their values:

Key	Description
Honeywell.UAS.Data.DC.DataSourceNames	Name of PI server to which Data logger is connected. NOTE More than one PI server can be added by separating each PI Server name with a ",".
Honeywell.UAS.Data.DC.SyncConfigFromUAS	Updating the SyncConfigFromUAS key value to True enables the Data Concentrator to automatically synchronize the History tag data source with Data Logger.
Honeywell.UAS.Data.DC.SyncConfigFromUAS_SyncHistoryDataSource	If the SyncHistoryDataSource key value is set to True , Data Concentrator will consider only the History Tag Source for synchronization. If the value is set to False Data Concentrator will considering only Read Tag Source for the synchronization of attribute to data logger tag table. NOTE Ensure SyncConfigFromUAS value is set to True for SyncHistoryDataSource to work.
Honeywell.UAS.Data.DC.SyncFromUAS	SyncFromUAS enables the Data Concentrator to automatically synchronize its tag definitions from APM. Update the key value to False .

4. Click **Save**.
5. For more information on how to configure Data Logger, refer **Data Logger User Guide**.

Manually Configuring Tags

If APM autosync is not enabled, you need to configure your tags so they point to the Asset Performance Management output (see [Tag Configuration](#)):

Name: Fully qualified Asset Performance Management

Attribute path: [Hierarchy Path].[Attribute Name]

APM Data Type

If you want to resolve the data type of data incoming from Asset Performance Management, the ValueDataType needs to contain one of the following options:

Identifier	DataType	Description
0	Double	Attempts to convert the incoming data to a Double.
1	Integer	Attempts to convert the incoming data to an Integer.
3	String	Attempts to convert the incoming data to a String.
4	AssetConnection	Attempts to convert the incoming data, which contains APM asset connection details, such as an asset path, to a String. Refer the Asset Performance Management documentation for more details on asset connections.
5	DateTime	Attempts to convert the incoming data to a DateTime.
-1	Default Unknown	Use to ignore datatype.

SQL

SQL entries must be configured in XML, and have the following configuration format:

```
<connectionxml>
  <sqlconnectionstring>
    Data Source=[DataSource];
    User=[UserName];
    Password=[Password]
  </sqlconnectionstring>
  <updatebatchsize>[BatchSize]</updatebatchsize>
  <utc>true</utc>
  <insertcommandtext>INSERT INTO [DatabaseName].[dbo].[TableName] VALUES
    (@AssetName, @TagName, @TimeStamp, @Value, @Quality, @UOM,
    @DataType)</insertcommandtext>
</connectionxml>
```

Where:

- **sqlconnectionstring**: The connection string to the SQL Server.
- **updatebatchsize**: The number of writes in a batch written to the SQL Server (default is 1000).
- **UTC**: Whether or not the system should treat incoming data as being UTC timestamped (if false, local system time zone is used).
- **insertcommandtext**: The insert command used to insert the historian data, either an insert statement or a stored procedure call. This command can accept any of the follow parameters:
 - @AssetName (varchar)
 - @TagName (varchar)
 - @TimeStamp (DateTime2)
 - @Value (varchar)
 - @Quality (int)
 - @UOM (varchar)
 - @DataType (int)

These are the fields the DC provides. Any other required fields should managed by your OLE RDI configuration.

XML Example:

```
"<connectionxml>
<sqlconnectionstring>
Data Source=MEMDataSource1;
User=DBUser1;
Password=Password1
</sqlconnectionstring>
<updatebatchsize>1000</updatebatchsize>
<utc>true</utc>
<insertcommandtext>INSERT INTO MEMDatabase.dbo.OutputTable VALUES (@AssetName,
@TagName, @TimeStamp, @Value, @Quality, @UOM, @DataType)</insertcommandtext>
</connectionxml>"
```

XML Stored Procedure Example:

```
"<connectionxml>
<sqlconnectionstring>
Data Source=MEMDataSource1;
User=DBUser1;
Password=Password1
</sqlconnectionstring>
```

```

<updatebatchsize>1000</updatebatchsize>

<utc>true</utc>

<insertcommandtext>EXEC InsertHistorianData @AssetName, @TagName, @TimeStamp,
@Value, @Quality, @UOM, @DataType</insertcommandtext>

</connectionxml>”

```

5.2.2 Tag Configuration

The Data Concentrator requires all tags to be processed should be defined in the tag configuration table.

The 'DLTag' table, contains the following columns:

1. **ID**: An automatically generated ID for the tag.
2. **Name**: The fully qualified historian tag name.

NOTE

The format of the tag name is dependent on the connected system. (For example, when using Asset Performance Management, the tag name must be the full attribute path, as shown in [Asset Performance Management](#).

3. **Alias**: The tag alias used in Datalogger sensor data files.
4. **Qualifier** (Optional): The qualifier to the alias that allows the full tag name to be resolved.
5. **FromUOM** (Optional): The **unit of measure** identifier for the source units. The spelling and case must match the contents of the unit conversion configuration file. Refer [Unit Conversions](#) for more information.
6. **ToUOM** (Optional): The unit of measure identifier for the destination units. The spelling and case must match the contents of the unit conversion configuration file. Refer [Unit Conversions](#) for more information..
7. **HistorianID**: The ID of the destination historian, refer [Historian Configuration](#) for more information..
8. **ValueDataType** (optional): The data type identifier, refer [Historian Configuration](#) for more information..

Example:

Assume that a sensor data file is produced with the following filename:

20160206_001147_751_M2X_Realtime.csv

where, **M2X** is the qualifier of the contained data.

If the contents of this file resemble the following:

0x99-1-1,0,192,06-02-2016 00:11:51.739

Then, a sample row in the DLTag table will be as follows:

- **Name**: Mem33:0x99-1-1:IP_VALUE
- **Alias**: 0x99-1-1
- **Qualifier**: M2X

- **FromUOM:** degF
- **ToUOM:** degC
- **HistorianID:** 1
- **ValueDataType:** -1

5.3 Unit Conversions

The Data Concentrator uses a unit conversion implementation called **Universal Unit Conversion** which operates on the principle that any conversion can be performed with the following formula:

$$\text{Converted Value} = A + B * \text{value} + C / (D + \text{value})$$

Thus, if the coefficients A, B, C and D are defined for a given source destination unit, it is possible to convert the value with the above formula.

The Data Concentrator includes a sample configuration file containing a large number of pre-defined unit conversions. The expected format of this file is CSV (comma separated values) with the following format:

FromUnit,ToUnit,A,B,C,D

This file can be modified at any time to add new conversion definitions or change existing conversion definitions.

NOTE

The Data Concentrator service will need to be restarted after any changes are made.

5.4 DC Counters

The Data Concentrator has a number of counters that can be used with other programs (for example OPC for Task Manager) to enter the data into OPC. It can then be used to write calculations and raise alarms (in the Alarm Viewer).

Example data flow: DC → OPC for Task Manager → ECM or APM OPC input → Calculation.

The following counters are available.

Counter	Description
DC Asset Instances	
DataAgeName	Data Age (seconds)
FileCountName	Processed Files
DetectedFileCountName	Detected Files
ArchivedFileCountName	Archived Files
BadFileCountName	Bad Files
BadRetriedFileCountName	Bad Files (Failed after maximum retries)
RecoverableErrorCountName	Recoverable Errors
UnrecoverableErrorCountName	Unrecoverable Errors
GatheringFileAverageSecondsName	Average Data Lag: 1-Gathering Time (seconds)
WaitingToProcessFileAverageSecondsName	Average Data Lag: 2-Waiting To Process Time (seconds)
ProcessingFileAverageSecondsName	Average Data Lag: 3-Processing Time (seconds)

Counter	Description
SendingDataAverageSecondsName	Average Data Lag: 4-Sending Time (seconds)
CompleteFileCountName	Average Data Lag: Averaged Files
OrphanedFileCountName	Average Data Lag: Orphaned Data Lag Records
GatheringFileLatestSecondsName	Latest Data Lag: 1-Gathering Time (seconds)
WaitingToProcessFileLatestSecondsName	Latest Data Lag: 2-Waiting To Process Time (seconds)
ProcessingFileLatestSecondsName	Latest Data Lag: 3-Processing Time (seconds)
SendingDataLatestSecondsName	Latest Data Lag: 4-Sending Time (seconds)
GatheringFileCurrentSecondsName	Current Data Lag: 1-Gathering Time (seconds)
WaitingToProcessFileCurrentSecondsName	Current Data Lag: 2-Waiting To Process Time (seconds)
ProcessingFileCurrentSecondsName	Current Data Lag: 3-Processing Time (seconds)
SendingDataCurrentSecondsName	Current Data Lag: 4-Sending Time (seconds)
GatheringFileCurrentCountName	Current File Count: 1-Gathering
WaitingToProcessFileCurrentCountName	Current File Count: 2-Waiting To Process
ProcessingFileCurrentCountName	Current File Count: 3-Processing
SendingDataCurrentCountName	Current File Count: 4-Sending
DC Overall	
InstanceCountName	Instances (Total)
ActiveInstanceCountName	Instances (Active)
Average[DCAssetCounterName]	Average of all the values for the DC counter.
Maximum[DCAssetCounterName]	Maximum of all the values for the DC counter.
Minimum[DCAssetCounterName]	Minimum of all the values for the DC counter.
Sum[DCAssetCounterName]	Sum of all the values for the DC counter.

5.4.1 Asset Performance Management - Data Concentrator File Processing Counter Example Usage

These counters are specific to an asset. There is one instance of these counter for each combination of asset/mode (real-time or historical) and input type (ae, log, other, sensor).

Example 1

The DC counter **DataAgeName** can be used in a calculation to detect if the value exceeds a sensible default. If your system is generating files every 30 seconds, it might make sense to warn if Data Age (seconds) exceeds 90 seconds (effectively 3 files behind).

This Data Age metric is specific to the scope of the DC. If you're outputting data to UFL and the PI Server is behind or down, that will not be reflected in this Data Age metric. However for other outputs that use synchronous writing to their output this Data Age metric is accurate.

Example 2

The DC counter **BadRetriedFileCountName** can be used in a calculation to detect if any files are ever rejected to the Bad folder due to the file exceeding the **MaxSuspectRecoverableErrorRetries** limit. Refer to "MaxSuspectRecoverableErrorRetries" section as to what this means.

Any change in the **BadRetriedFileCountName** counter usually means that there is either a file that cannot be parsed or a DC output has crashed/seized in such a way that it appears to still be functional, but it is failing to work.

Example 3

The **Latest Data Lag** DC Counters can be used in calculations to detect if any files have gotten stuck within the processing of the DC:

1. **GatheringFileLatestSecondsName**: Growth in this counter can indicate an issue with the WebDav/SFTP server or network throughput issues.
2. **WaitingToProcessFileLatestSecondsName**: Growth in this counter can indicate something is causing the DC to be unable to open and process files optimally, possibly file system issues (very full or busy drives) or file locking/permissions issues.
3. **ProcessingFileLatestSecondsName**: Growth in this counter can indicate trouble processing files themselves, for example, a maxed CPU can cause this.
4. **SendingDataLatestSecondsName**: Growth in this counter can indicate throughput issues to the output systems. Investigate the connectivity and throughputs to the output systems.

Example 4

The **Average Data Lag** DC counters (**GatheringFileAverageSecondsName**, **WaitingToProcessFileAverageSecondsName**, **ProcessingFileAverageSecondsName**, **SendingDataAverageSecondsName**, **CompleteFileCountName**, and **OrphanedFileCountName**) can be used in calculations to detect if any of the stages of processing of the DC are performing slowly.

Comparing these values against thresholds established for a given install (every install is different, but monitoring these values to find a sensible threshold for a given site) and then using that threshold to detect any major deviations can help to identify performance issues early before a large backlog of data is created.

Example 5

The DC counter **UnrecoverableErrorCountName** can be used in a calculation to detect if any files are ever rejected to the Bad folder due to an unrecoverable error. In a running system this should never happen and indicates a possible issue in the data flow.

5.4.2 Asset Performance Management - Data Concentrator File Processing (Aggregated) Counters Example Usage

These counters are an aggregation of all assets within the DC. There is one instance of these counter for each input type (ae, log, other, sensor).

Example 1

The **Average - Average Data Lag** DC counters can be used in calculations to detect if the DC is capable of keeping up to the data flow. Summing up the **ProcessingFileAverageSecondsName** and **SendingDataAverageSecondsName** counter values can represent the full cycle time needed to handle a file, averaged over the DC.

For example, if the sum of these values is 1 second, and the DC is configured to use 4 threads, then the DC is capable of handling 4 file every second. If the install has 100 assets each generating a sensor file every 30 seconds, then the DC needs to process over 3 (100 assets/30 seconds) files per seconds to keep up.

NOTE

For the UFL output, only the `ProcessingFileAverageSecondsName` counter is needed for the above metric, as UFL sending is an asynchronous process (the UFL server loads our output files at its own pace, and does not delay the DC processing).

However, creating the same calculation to monitor only the `SendingDataAverageSecondsName` counter for a UFL output gives you an indication of whether the PI server is capable of keeping up to the data flow.

Example 2

The **Maximum** counters can be useful to see if any of the asset's counters have exceeded a threshold. This can help works as a first line of defense against exceeding performance thresholds.

Example 3

Comparing the **InstanceCountName** and **ActiveInstanceCountName** counters can be useful to see if any of the assets have stopped reporting their counters values.

This may not indicate a problem with the operation of the DC processing files itself (the DC can operate without counters enabled), but this should never happen and would indicate an internal issue in the DC.

Example 4

The **Sum** counters can be useful to see if any absolute thresholds have been breached. If your running DC expects exactly 0 Unrecoverable Errors, then using the Sum - Unrecoverable Errors counter will ensure that your calculation raises an alarm if even one is detected.

DATA CONCENTRATOR FORWARDING AGENT SERVICE CONFIGURATION (OPTIONAL)

6.1 Forwarding Agent Configuration File

All Data Concentrator Forwarding Agent configurations are stored in the below location:

```
<Install Location>:\Program Files (x86)\Matrikon\Data Concentrator Forwarding Agent\Matrikon.Datalogger.ForwardAgent.Service.exe.config
```

NOTE

For any configuration changes to take effect, the Matrikon Data Concentrator Forwarding Agent service must be restarted.

6.1.1 Global Configuration

Under the **<applicationSettings>** section of the configuration file are a number of configuration settings that are global to the Data Concentrator Forwarding Agent.

NOTE

Certain reserved XML characters (such as angle brackets) need to be suitably escaped.

For example, '<' needs to be represented as '<'.

Parameter	Description	Example
TransferExtension	The temporary file extension to use when transferring a file. After successful transfer, the file will be named back remotely to its original name. If a blank TransferExtension is specified, the file will be transferred directly to the proper file name.	. xfer
CompressFiles	If True , the Forwarding Agent will gzip uncompressed files before transferring them to the remote server.	False
ProtectCredentials	If True , the SFTPUser and SFTPPassword parameters will be automatically encrypted in the configuration file when the Forwarding Agent starts. It is recommended that this is left to 'False' until the system has been	False

Parameter	Description	Example
	shown to be in a working state.	
LockRetryMS	The time (in milliseconds) that the Forwarding Agent waits to retry opening a file if the file is locked (for example, by another application).	10
MaxLockRetries	The number of times the Forwarding Agent will retry opening a file if the file is locked (for example, by another application). The maximum amount of time (in ms) the Forwarding Agent will try to open a locked file before giving up is: $\text{LockRetryMS} * \text{MaxLockRetries}$	1000
ConnectionRetryMS	The time in milliseconds that the Forwarding Agent will wait before trying to re-establish its connection to the SFTP server in the event of a network error.	5000
FileTimestampPattern	A regular expression identifying the timestamp portion in a filename.	<code>^....._....._...</code>
FileTimestampFormat	The format of the file timestamp, using .NET date format tokens. Please see Appendix A for a list of .NET date format tokens.	<code>yyyyMMdd_HH:mm:ss_fff</code>
EnableHeartbeat	If set to true, information about the health of the Forwarding Agent is made available to external processes. This option should be set to False .	False
LimitThreads	Specify False if the Forwarding Agent should use a thread for processing each directory set. Specify 'True' to use a capped thread pool for processing.	False
MaxThreads	If LimitThreads is enabled, specifies the maximum number of threads that can run simultaneously.	32
InstanceName	This is the name of this installed forwarding agent. Used in the counter instance names for the aggregated system data health counters. Each install within one deployment must be unique.	FA1
AggregateCounterEvaluationIntervalSeconds	This is the interval (in seconds) that the system will evaluate the aggregated system data health counters, defaulted to 0. 0 means disabled.	0
SDHFileCacheMaximumAgeSeconds	This is the size of the cache (in seconds)	300

Parameter	Description	Example
	that the system will maintain for calculating the aggregated system data health counters, defaulted to 300.	
SDHEvaluationIntervalSeconds	This is the interval in seconds that the system will evaluate the system data health counters, defaulted to 0. 0 means disabled.	0

6.1.2 Transferer Configuration

One or more transfer connections can be defined to send data to a remote destination. The section covers the available transfer methods.

FileCopy

To define a FileCopy connection, add an entry to the **transferers/fileCopyTransferers** section with the format:

```
<add id="id" maxConcurrency="10" rootDirectory="C:\copy\"/>
```

Where:

- **id**: A unique identifier for this connection. This id will be referenced as a **destination** during the DirectorySet configuration. Refer [Directory Set Configuration](#) for more information.
- **maxConcurrency**: Specifies how many simultaneous transfers will be attempted with this connection.
- **rootDirectory**: The root folder (local or remote shared directory) that the files will be copied to. Security for this transferer is managed by Windows folder security.

Example:

```
<transferers default="fileCopy1">
  <sftpTransferers>
    <add id="fileCopy1" maxConcurrency="10" rootDirectory="C:\copy\" />
  </sftpTransferers>
</transferers>
```

NOTE

When configuring the remoteDestinationLocation section of directorySets/directorySet/directory (Refer [Directory Set Configuration](#) for more information) for a FileCopy transferer, only prefix the content with a "\" or "/" character if you intend remoteDestinationLocation to be an absolute path.

If the remoteDestinationLocation is not prefixed with a "\" or "/" then it will be appended to the rootDirectory path to construct the full file copy folder path.

SFTP

To define an SFTP connection, add an entry to the **transferers/sftpTransferers** section with the format:

```
<add id="id" host="host" user="user" password="password" port="port" />
```

Where:

- **id**: A unique identifier for this connection. This id will be referenced as a 'destination' during the Directory Set Configuration. Refer [Directory Set Configuration](#) for more information.
- **host**: A host name or IP address of the destination SFTP server.
- **user**: The user to use for the SFTP connection. This field may reference a protected setting, refer [Protected Settings](#) for more information.
- **password**: The user password to use for the SFTP connection. This field may reference a protected setting refer [Protected Settings](#) for more information.
- **port**: The port to use for the SFTP connection (typically 22).

Example:

```
<transferers default="default">
<sftpTransferers>
<add id="default" host="host1" user="user1" password="pword1" port="22" />
</sftpTransferers>
</transferers>
```

WebDAV

To define a WebDAV connection, add an entry to the **transferers/webDAVTransferers** section with the format:

```
<add id="id" rootUri="rootUri" user="user" password="password1 domain="domain"
maxConcurrency="maxConcurrency"
clientCertificateThumbprint="clientCertificateThumbprint" />
```

Where:

- **id**: A unique identifier for this connection. This id will be referenced as a **destination** during the DirectorySet configuration.
- **rootUri**: The web address representing the root of the destination location. The 'remoteDestinationLocation' for each directory set (Refer [Directory Set Configuration](#) for more information) will be relative to this location.
Either http or https can be used for the rootUri. If https is used, enter the fully qualified server name matching the server's certificate.
- **user**: The user to use for the web connection. This field may reference a protected setting, refer [Protected Settings](#) for more information.
If specified, the user/password/domain settings will form Basic HTTP credentials to the destination web server.
If not specified, the default network credentials of the Forwarding Agent service process will be used, if Integrated Windows Authentication is enabled on the destination web server.
- **password**: The user password to use for the web connection. This field may reference a protected setting, refer [Protected Settings](#) for more information
- **domain**: The user domain to use for the web connection. This field may reference a protected setting, refer [Protected Settings](#) for more information

- **maxConcurrency**: Specifies how many simultaneous transfers will be attempted to this connection.
- **clientCertificateThumbprint**: If specified, uses the identified client X509 certificate for the connection requests (for increased authentication security). The referenced certificate must be present in the Forwarding Agent Service user's Personal certificate store.

Example:

```
<transferers default="default">

<webDAVTransferers>

<add id="default" rootUri="http://host/webroot" user="user1"
password="password1" domain="" maxConcurrency="2"
clientCertificateThumbprint="622b18fbbdbf867edb48df373263a6b68e05cae1" />

</webDAVTransferers>

</transferers>
```

Refer [WebDAV Support Material](#) for additional documentation and best-practices for configuring and securing WebDAV.

6.1.3 Protected Settings

The **<protectedSettings>** section allows sensitive information to be stored as key/value pairs which can then be encrypted and referenced from other configuration locations. Securing configuration data consists of the following steps:

Add key/value pairs to the **<protectedSettings>** section as required. Each key/value pair takes the following form:

```
<setting name="key" serializeAs="String">

<value>value</value>

</setting>
```

The following example shows two passwords stored in the **<protectedSettings>** section:

```
<protectedSettings> <
setting name="ProdPassword" serializeAs="String">
<value>secret1</value>
</setting> <setting name="DevPassword" serializeAs="String">
<value>secret2</value>
</setting></protectedSettings>
```

Reference Protected Configuration Values

User and Password fields in the transferers configuration can reference settings from the **<protectedSettings>** section. To do this, enclose the setting keys in braces **{}**.

The following example defines an SFTP transferer, referencing the 'DevPassword' protected setting:

```

<transferers default="default">

<sftpTransferers>

<add id="default" host="host1" user="user1" password="{DevPassword}" port="22"
/>

</sftpTransferers>

</transferers>

```

Configure the System to Encrypt the Protected Settings Section

After defining the required protected key/value pairs, change the '**ProtectCredentials**' to **True**.

```

<setting name="ProtectCredentials" serializeAs="String">

<value>True</value>

</setting>

```

After restarting the Forwarding Agent service, the <protectedSettings> section will be encrypted.

Data Protection Method

By default, the system will use DPAPI at the machine-level to secure the contents of the <protectedSettings> section, meaning that the section can be decrypted by users with access to machine.

To use DPAPI at the user-level (meaning that only the service user that encrypted the data can read it), configure the Data Concentrator Forwarding Agent service to run as a specific user and define the following protection provider (commented out in the default configuration file) before securing the <protectedSettings> section:

```

<configProtectedData defaultProvider="DPAPIProtection">

<providers>

<add name="DPAPIProtection"

type="System.Configuration.DpapiProtectedConfigurationProvider, System.Configuration,
Version=2.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a"

useMachineProtection="false" />

</providers>

</configProtectedData>

```

6.1.4 Directory Set Configuration

Any number of Directory Sets can be configured, each having a list of directories with configurable properties. The processing of files within a Directory Set is serialized, with relative directory priorities definable within the set (similar to the Data Concentrator's Alarm/Sensor/Log priority system).

In the <**directorySets**> section are the specific configuration settings for each Directory Set, which in most cases will have direct correspondence to a single Datalogger archive directories. A Directory Set definition (for a fictitious 'M2X' Datalogger) resembles the following:

```

<directorySet ID="M2X">

```

```
<directory>
<localSourceDirectory>C:\watch\Alarm</localSourceDirectory>
<remoteDestinationLocation>/M2X/Alarm</remoteDestinationLocation>
<localArchiveDirectory>C:\Archive\M2X\Alarm</localArchiveDirectory>
<priority>1</priority>
<filter>*.pgd</filter>
</directory>
<directory>
<localSourceDirectory>C:\watch\Sensor</localSourceDirectory>
<remoteDestinationLocation>/M2X/Sensor</remoteDestinationLocation>
<localArchiveDirectory>C:\Archive\M2X\Sensor</localArchiveDirectory>
<priority>2</priority>
<filter>*.bin</filter>
</directory>
<directory>
<localSourceDirectory>C:\watch\Log</localSourceDirectory>
<remoteDestinationLocation>/M2X/Log</remoteDestinationLocation>
<localArchiveDirectory>C:\Archive\M2X\Log</localArchiveDirectory>
<priority>3</priority>
<filter>*.log</filter>
</directory>
</directorySet>
```

Each **directorySet** entry has the following configuration options:

Parameter	Description	Example
ID	A unique identifier for the directorySet. NOTE This value is currently not used by the Forwarding Agent, but should be set to a sensible value to improve the maintainability of the configuration.	M2X
directory/localSourceDirectory	The source directory that is monitored for files, in this directory entry.	C:\watch\Sensor
directory/remoteDestinationLocation	The destination location on the SFTP	/M2X/Alarm

Parameter	Description	Example
	server for files in this directory entry.	
directory/localArchiveDirectory	The location on the source machine where files are moved after successful transfers.	C:\Datalogger\Archive\Sensor
directory/priority	The priority of files within this directory relative to files in other directories in this set. The lower the number, the higher the priority.	1
directory/filter	The filter affecting the types of files monitored in the directory.	*.bin

6.1.5 File Ordering

For each directory, if a **timestamp pattern** is specified, the files are internally ordered **chronologically** using the timestamps extracted from the filenames using the **timestamp pattern** and **timestamp format** configured for that input location.

If a **blank** timestamp pattern is specified for a directory, alpha-numeric ordering will be used. The following example shows alpha-numeric ordering used for the first directory, and chronological ordering used for the second directory (assuming a valid global FileTimestampPattern and FileTimestampFormat) have been configured.

```
<directorySet ID="M2X">
  <directory>
    <localSourceDirectory>C:\watch\Alarm</localSourceDirectory>
    <remoteDestinationLocation>/M2X/Alarm</remoteDestinationLocation>
    <localArchiveDirectory>C:\Archive\M2X\Alarm</localArchiveDirectory>
    <priority>1</priority>
    <filter>*.pgd</filter>
    <fileTimestampPattern></fileTimestampPattern>
  </directory>
  <directory>
    <localSourceDirectory>C:\watch\Sensor</localSourceDirectory>
    <remoteDestinationLocation>/M2X/Sensor</remoteDestinationLocation>
    <localArchiveDirectory>C:\Archive\M2X\Sensor</localArchiveDirectory>
    <priority>2</priority>
    <filter>*.bin</filter>
  </directory>
</directorySet>
```

6.1.6 Logging

The Data Concentrator Forwarding Agent uses Apache log4net for logging, and supports the full range of log4net configuration parameters.

Documentation on the log4net configuration can be found in the below link:

<http://logging.apache.org/log4net/release/config-examples.html>

By default, the Data Concentrator Forwarding Agent uses the following basic configuration defined in the **<log4net>** section:

```
<log4net>
  <appender name="RollingFileAppender" type="log4net.Appender.RollingFileAppender">
    <param name="File" value="C:/ForwardingAgentLog.txt" />
    <param name="AppendToFile" value="true" />
    <param name="RollingStyle" value="Size" />
    <param name="MaxSizeRollBackups" value="10" />
    <param name="MaximumFileSize" value="1024KB" />
    <param name="StaticLogFileName" value="true" />
    <layout type="log4net.Layout.PatternLayout">
      <param name="ConversionPattern" value="%d [%t] %-5p %c - %m%n" />
    </layout>
  </appender>
  <root>
    <level value="DEBUG" />
    <appender-ref ref="RollingFileAppender" />
  </root>
</log4net>
```

The **RollingFileAppender** will maintain a fixed set of log files (defined by the **MaxSizeRollBackups** and **MaximumFileSize** parameters), with a base **file** name defined by the **File** parameter. This configuration should be suitable for most situations.

The log **level** should also be defined as appropriate for your stage of the project. When first commissioning the system, leaving the level at **DEBUG** will make debugging problems easier, at the cost of a large amount of logging statements. In general production, a log level of **INFO** would be appropriate.

6.2 FA Counters

The Data Concentrator Forwarding Agent has a number of counters that can be used with other programs (for example OPC for Task Manager) to enter the data into OPC. It can then be used to write calculations and raise alarms (in the Alarm Viewer).

Example data flow: FA → OPC for Task Manager → ECM or APM OPC input → Calculation.

Following are the counters available:

Counter	Description
FA Asset Instances	
DetectedFileCountName	Detected Files
ArchivedFileCountName	Archived Files
ForwardedFileCountName	Forwarded Files
CompressedFileCountName	Compressed Files
UnrecoverableErrorCountName	Unrecoverable Errors
GatheringFileAverageSecondsName	Average Data Lag: 1-Gathering Time (seconds)
WaitingToSendFileAverageSecondsName	Average Data Lag: 2-Waiting To Send Time (seconds)
SendingDataAverageSecondsName	Average Data Lag: 3-Sending Time (seconds)
CompleteFileCountName	Average Data Lag: Averaged Files
OrphanedFileCountName	Average Data Lag: Orphaned Data Lag Records
GatheringFileLatestSecondsName	Latest Data Lag: 1-Gathering Time (seconds)
WaitingToSendFileLatestSecondsName	Latest Data Lag: 2-Waiting To Send Time (seconds)
SendingDataLatestSecondsName	Latest Data Lag: 3-Sending Time (seconds)
GatheringFileCurrentSecondsName	Current Data Lag: 1-Gathering Time (seconds)
WaitingToSendFileCurrentSecondsName	Current Data Lag: 2-Waiting To Send Time (seconds)
SendingDataCurrentSecondsName	Current Data Lag: 3-Sending Time (seconds)
GatheringFileCurrentCountName	Current File Count: 1-Gathering
WaitingToSendFileCurrentCountName	Current File Count: 2-Waiting To Send
SendingDataCurrentCountName	Current File Count: 3-Sending
FA Overall	
InstanceCountName	Instances (Total)
ActiveInstanceCountName	Instances (Active)
Average[FAAssetCounterName]	Average of all the values for the FA counter.
Maximum[FAAssetCounterName]	Maximum of all the values for the FA counter.
Minimum[FAAssetCounterName]	Minimum of all the values for the FA counter.
Sum[FAAssetCounterName]	Sum of all the values for the FA counter.

For examples of ways to use these counters, see "[Asset Performance Management - Data Concentrator File Processing Counter Example Usage](#)" and "[Asset Performance Management - Data Concentrator File Processing \(Aggregated\) Counters Example Usage](#)".

DATA CONCENTRATOR ARCHIVE MAINTAINER CONFIGURATION (OPTIONAL)

7.1 Archive Maintainer Configuration File

All Data Concentrator Forwarding Agent configuration is stored in the below location:

<Install Location>:\Program Files (x86)\Matrikon\Data Concentrator Archive Maintainer\Matrikon.Datalogger.ArchiveMaintainer.exe.config

7.1.1 Global Configuration

Under the **<applicationSettings>** section of the configuration file are a number of configuration settings that serve as defaults for the Archive Maintainer.

NOTE

Certain reserved XML characters (such as angle brackets) need to be suitably escaped.

For example, '<' needs to be represented as '<'.

Parameter	Description	Example
ArchiveMaintainerIntervalMinutes	The interval to auto-run the archive maintainer processing against the active data concentrator "Archive" folders. 0 = disabled	0
ArchiveMaintainerMaxDirectorySizeBytes	The (default) maximum allowed size of a directory in bytes. If the total size of files in a directory exceeds this amount, files are deleted (from oldest to newest) until the directory size is within limits. To disable directory size limits, specify 0 for this value.	104857600 (100 megabytes)
ArchiveMaintainerOldestFileAgeDays	The (default) maximum allowed age of a file. Files older than this value are deleted. To disable file age limits, specify 0 for this value.	365
ArchiveMaintainerMaxFileCount	The (default) maximum number of files to allow within a directory. If the total number of files within a directory exceeds this amount, files are deleted (from oldest to newest) until the number of files within the directory is within limits. To disable file count limits, specify 0 for this value.	1000
ArchiveMaintainerFilters	List of file type to exclude from the archive.	*.gz, *.bin, *.csv
ArchiveMaintainerRecursive	Whether or not the archive procession is recursive.	True

7.1.2 Maintained Directory Configuration

Any number of Maintained Directories can be configured, each having any combination of maintenance rules.

In the **<maintainedDirectories>** section are the specific configuration settings for each directory. A Maintained Directory definition resembles the following:

```
<maintainedDirectories>
<maintainedDirectory>
<directory>c:\M2X\Sensor</directory>
<maxFileCount>1000</maxFileCount>
<maxDirectorySizeBytes>104857600</maxDirectorySizeBytes>
<oldestFileAgeDays>365</oldestFileAgeDays>
</maintainedDirectory>
</maintainedDirectories>
```

Each maintainedDirectory entry has the following configuration options:

Parameter	Description	Example
directory	The absolute path to the directory to maintain.	C:\M2X\Sensor
maxFileCount	The maximum number of files to allow within a directory. If the total number of files within a directory exceeds this amount, files are deleted (from oldest to newest) until the number of files within the directory is within limits. To disable file count limits, specify 0 for this value. If this parameter is omitted, the MaxFileCount specified in the global configuration will be used. If this parameter is specified, it will override the MaxFileCount specified in the global configuration for this directory.	1000
maxDirectorySizeBytes	The maximum allowed size of a directory in bytes. If the total size of files in a directory exceeds this amount, files are deleted (from oldest to newest) until the directory size is within limits. To disable directory size limits, specify 0 for this value. If this parameter is omitted, the MaxDirectorySizeBytes specified in the global configuration will be used. If this parameter is specified, it will override the MaxDirectorySizeBytes specified in the global configuration for this directory.	104857600
oldestFileAgeDays	The maximum allowed age of a file. Files older than this value are deleted. To disable file age limits, specify 0 for this value. If this parameter is omitted, the OldestFileAgeDays specified in the global configuration will be used. If this parameter is specified, it will override the OldestFileAgeDays specified	

Parameter	Description	Example
	in the global configuration for this directory.	

7.1.3 Logging

The Data Concentrator Archive Maintainer uses Apache log4net for logging, and supports the full range of log4net configuration parameters.

Documentation on the log4net configuration can be found in the below link:

<http://logging.apache.org/log4net/release/config-examples.html>

By default, the Data Concentrator Archive Maintainer uses the following basic configuration defined in the `<log4net>` section:

```
<log4net>

  <appender name="RollingFileAppender" type="log4net.Appender.RollingFileAppender">
    <param name="File" value="C:/ArchiveMaintLog.txt" />
    <param name="AppendToFile" value="true" />
    <param name="RollingStyle" value="Size" />
    <param name="MaxSizeRollBackups" value="10" />
    <param name="MaximumFileSize" value="1024KB" />
    <param name="StaticLogFileName" value="true" />
    <layout type="log4net.Layout.PatternLayout">
      <param name="ConversionPattern" value="%d [%t] %-5p %c - %m%n" />
    </layout>
  </appender>

  <root>
    <level value="DEBUG" />
    <appender-ref ref="RollingFileAppender" />
  </root>

</log4net>
```

The **RollingFileAppender** will maintain a fixed set of log files (defined by the **MaxSizeRollBackups** and **MaximumFileSize** parameters), with a base file name defined by the **File** parameter. This configuration should be suitable for most situations.

The log **level** should also be defined as appropriate for your stage of the project. When first commissioning the system, leaving the level at **DEBUG** will make debugging problems easier, at the cost of a large amount of logging statements. In general production, a log level of **INFO** would be appropriate.

A

.NET DATE FORMAT TOKENS

Format pattern	Description
d, %d	The day of the month. Single-digit days do not have a leading zero. The application specifies "%d" if the format pattern is not combined with other format patterns.
dd	The day of the month. Single-digit days have a leading zero.
ddd	The abbreviated name of the day of the week, as defined in AbbreviatedDayNames .
dddd	The full name of the day of the week, as defined in DayNames .
f, %f	The fraction of a second in single-digit precision. The remaining digits are truncated. The application specifies "%f" if the format pattern is not combined with other format patterns.
ff	The fraction of a second in double-digit precision. The remaining digits are truncated.
fff	The fraction of a second in three-digit precision. The remaining digits are truncated.
ffff	The fraction of a second in four-digit precision. The remaining digits are truncated.
fffff	The fraction of a second in five-digit precision. The remaining digits are truncated.
ffffff	The fraction of a second in six-digit precision. The remaining digits are truncated.
fffffff	The fraction of a second in seven-digit precision. The remaining digits are truncated.
F, %F	Displays the most significant digit of the seconds fraction. Nothing is displayed if the digit is zero. The application specifies "%F" if the format pattern is not combined with other format patterns.
FF	Displays the two most significant digits of the seconds fraction. However, trailing zeros, or two zero digits, are not displayed.
FFF	Displays the three most significant digits of the seconds fraction. However, trailing zeros, or three zero digits, are not displayed.
FFFF	Displays the four most significant digits of the seconds fraction. However, trailing zeros, or four zero digits, are not displayed.
FFFFF	Displays the five most significant digits of the seconds fraction. However, trailing zeros, or five zero digits, are not displayed.
FFFFFF	Displays the six most significant digits of the seconds fraction. However, trailing zeros, or six zero digits, are not displayed.
FFFFFFF	Displays the seven most significant digits of the seconds fraction. However, trailing zeros, or seven zero digits, are not displayed.
gg	The period or era. This pattern is ignored if the date to be formatted does not have an associated period or era string.
h, %h	The hour in a 12-hour clock. Single-digit hours do not have a leading zero. The application specifies "%h" if the format pattern is not combined with other format patterns.

Format pattern	Description
hh	The hour in a 12-hour clock. Single-digit hours have a leading zero.
H, %H	The hour in a 24-hour clock. Single-digit hours do not have a leading zero. The application specifies "%H" if the format pattern is not combined with other format patterns.
HH	The hour in a 24-hour clock. Single-digit hours have a leading zero.
K	Different values of the Kind property, that is, Local, Utc, or Unspecified.
m, %m	The minute. Single-digit minutes do not have a leading zero. The application specifies "%m" if the format pattern is not combined with other format patterns.
mm	The minute. Single-digit minutes have a leading zero.
M, %M	The numeric month. Single-digit months do not have a leading zero. The application specifies "%M" if the format pattern is not combined with other format patterns.
MM	The numeric month. Single-digit months have a leading zero.
MMM	The abbreviated name of the month, as defined in AbbreviatedMonthNames .
MMMM	The full name of the month, as defined in MonthNames .
s, %s	The second. Single-digit seconds do not have a leading zero. The application specifies "%s" if the format pattern is not combined with other format patterns.
ss	The second. Single-digit seconds have a leading zero.
t, %t	The first character in the AM/PM designator defined in AMDesignator or PMDesignator , if any. The application specifies "%t" if the format pattern is not combined with other format patterns.
tt	The AM/PM designator defined in AMDesignator or PMDesignator , if any. Your application should use this format pattern for languages for which it is necessary to maintain the distinction between AM and PM. An example is Japanese, for which the AM and PM designators differ in the second character instead of the first character.
y, %y	The year without the century. If the year without the century is less than 10, the year is displayed with no leading zero. The application specifies "%y" if the format pattern is not combined with other format patterns.
yy	The year without the century. If the year without the century is less than 10, the year is displayed with a leading zero.
yyy	The year in three digits. If the year is less than 100, the year is displayed with a leading zero.
yyyy	The year in four or five digits (depending on the calendar used), including the century. Pads with leading zeros to get four digits. Thai Buddhist and Korean calendars have five-digit years. Users selecting the "yyyy" pattern see all five digits without leading zeros for calendars that have five digits. Exception: the Japanese and Taiwan calendars always behave as if "yy" is selected.
yyyyy	The year in five digits. Pads with leading zeros to get five digits. Exception: the Japanese and Taiwan calendars always behave as if "yy" is selected.
yyyyyy	The year in six digits. Pads with leading zeros to get six digits. Exception: the Japanese and Taiwan calendars always behave as if "yy" is selected. The pattern can be continued with a longer string of "y"s padding with more leading zeros.
z, %z	The time zone offset ("+" or "-" followed by the hour only). Single-digit hours do not have a leading zero. For example, Pacific Standard Time is "-8". The application specifies "%z" if the format pattern is not combined with other format patterns.

Format pattern	Description
zz	The time zone offset ("+" or "-" followed by the hour only). Single-digit hours have a leading zero. For example, Pacific Standard Time is "-08".
zzz	The full time zone offset ("+" or "-" followed by the hour and minutes). Single-digit hours and minutes have leading zeros. For example, Pacific Standard Time is "-08:00".
:	The default time separator defined in TimeSeparator .
/	The default date separator defined in DateSeparator .
% c	Where c is a format pattern if used alone. To use format pattern "d", "f", "F", "h", "m", "s", "t", "y", "z", "H", or "M" by itself, the application specifies "%d", "%f", "%F", "%h", "%m", "%s", "%t", "%y", "%z", "%H", or "%M". The "%" character can be omitted if the format pattern is combined with literal characters or other format patterns.
\ c	Where c is any character. Displays the character literally. To display the backslash character, the application should use "\\".

B

SUPPORTED TIME ZONES

All time zones supported by Windows will be available for use in the Data Concentrator. The following table lists some examples of the time zones, listed by ID.

To see the time zones available on your system, view the folders listed under the following registry key item:

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Time Zones

ATTENTION

Changing registry settings can create problems with the operating system. This key should be used to confirm the time zone settings are supported in the system, but there is no need for the data to be altered.

Id	Display Name
Morocco Standard Time	(GMT) Casablanca
GMT Standard Time	(GMT) Greenwich Mean Time : Dublin Edinburgh Lisbon London
Greenwich Standard Time	(GMT) Monrovia Reykjavik
W. Europe Standard Time	(GMT+01:00) Amsterdam Berlin Bern Rome Stockholm Vienna
Central Europe Standard Time	(GMT+01:00) Belgrade Bratislava Budapest Ljubljana Prague
Romance Standard Time	(GMT+01:00) Brussels Copenhagen Madrid Paris
Central European Standard Time	(GMT+01:00) Sarajevo Skopje Warsaw Zagreb
W. Central Africa Standard Time	(GMT+01:00) West Central Africa
Jordan Standard Time	(GMT+02:00) Amman
GTB Standard Time	(GMT+02:00) Athens Bucharest Istanbul
Middle East Standard Time	(GMT+02:00) Beirut
Egypt Standard Time	(GMT+02:00) Cairo
South Africa Standard Time	(GMT+02:00) Harare Pretoria
FLE Standard Time	(GMT+02:00) Helsinki Kyiv Riga Sofia Tallinn Vilnius
Israel Standard Time	(GMT+02:00) Jerusalem
E. Europe Standard Time	(GMT+02:00) Minsk
Namibia Standard Time	(GMT+02:00) Windhoek
Arabic Standard Time	(GMT+03:00) Baghdad
Arab Standard Time	(GMT+03:00) Kuwait Riyadh
Russian Standard Time	(GMT+03:00) Moscow St. Petersburg Volgograd

Id	Display Name
E. Africa Standard Time	(GMT+03:00) Nairobi
Georgian Standard Time	(GMT+03:00) Tbilisi
Iran Standard Time	(GMT+03:30) Tehran
Arabian Standard Time	(GMT+04:00) Abu Dhabi Muscat
Azerbaijan Standard Time	(GMT+04:00) Baku
Caucasus Standard Time	(GMT+04:00) Caucasus Standard Time
Mauritius Standard Time	(GMT+04:00) Port Louis
Armenian Standard Time	(GMT+04:00) Yerevan
Afghanistan Standard Time	(GMT+04:30) Kabul
Ekaterinburg Standard Time	(GMT+05:00) Ekaterinburg
Pakistan Standard Time	(GMT+05:00) Islamabad Karachi
West Asia Standard Time	(GMT+05:00) Tashkent
India Standard Time	(GMT+05:30) Chennai Kolkata Mumbai New Delhi
Sri Lanka Standard Time	(GMT+05:30) Sri Jayawardenepura
Nepal Standard Time	(GMT+05:45) Kathmandu
N. Central Asia Standard Time	(GMT+06:00) Almaty Novosibirsk
Central Asia Standard Time	(GMT+06:00) Astana Dhaka
Myanmar Standard Time	(GMT+06:30) Yangon (Rangoon)
SE Asia Standard Time	(GMT+07:00) Bangkok Hanoi Jakarta
North Asia Standard Time	(GMT+07:00) Krasnoyarsk
China Standard Time	(GMT+08:00) Beijing Chongqing Hong Kong Urumqi
North Asia East Standard Time	(GMT+08:00) Irkutsk Ulaan Bataar
Singapore Standard Time	(GMT+08:00) Kuala Lumpur Singapore
W. Australia Standard Time	(GMT+08:00) Perth
Taipei Standard Time	(GMT+08:00) Taipei
Tokyo Standard Time	(GMT+09:00) Osaka Sapporo Tokyo
Korea Standard Time	(GMT+09:00) Seoul
Yakutsk Standard Time	(GMT+09:00) Yakutsk
Cen. Australia Standard Time	(GMT+09:30) Adelaide
AUS Central Standard Time	(GMT+09:30) Darwin
E. Australia Standard Time	(GMT+10:00) Brisbane
AUS Eastern Standard Time	(GMT+10:00) Canberra Melbourne Sydney
West Pacific Standard Time	(GMT+10:00) Guam Port Moresby

Id	Display Name
Tasmania Standard Time	(GMT+10:00) Hobart
Vladivostok Standard Time	(GMT+10:00) Vladivostok
Central Pacific Standard Time	(GMT+11:00) Magadan Solomon Is. New Caledonia
New Zealand Standard Time	(GMT+12:00) Auckland Wellington
Fiji Standard Time	(GMT+12:00) Fiji Kamchatka Marshall Is.
Tonga Standard Time	(GMT+13:00) Nuku'alofa
Azores Standard Time	(GMT-01:00) Azores
Cape Verde Standard Time	(GMT-01:00) Cape Verde Is.
Mid-Atlantic Standard Time	(GMT-02:00) Mid-Atlantic
E. South America Standard Time	(GMT-03:00) Brasilia
Argentina Standard Time	(GMT-03:00) Buenos Aires
SA Eastern Standard Time	(GMT-03:00) Georgetown
Greenland Standard Time	(GMT-03:00) Greenland
Montevideo Standard Time	(GMT-03:00) Montevideo
Newfoundland Standard Time	(GMT-03:30) Newfoundland
Atlantic Standard Time	(GMT-04:00) Atlantic Time (Canada)
SA Western Standard Time	(GMT-04:00) La Paz
Central Brazilian Standard Time	(GMT-04:00) Manaus
Pacific SA Standard Time	(GMT-04:00) Santiago
Venezuela Standard Time	(GMT-04:30) Caracas
SA Pacific Standard Time	(GMT-05:00) Bogota Lima Quito Rio Branco
Eastern Standard Time	(GMT-05:00) Eastern Time (US & Canada)
US Eastern Standard Time	(GMT-05:00) Indiana (East)
Central America Standard Time	(GMT-06:00) Central America
Central Standard Time	(GMT-06:00) Central Time (US & Canada)
Central Standard Time (Mexico)	(GMT-06:00) Guadalajara Mexico City Monterrey - New
Mexico Standard Time	(GMT-06:00) Guadalajara Mexico City Monterrey - Old
Canada Central Standard Time	(GMT-06:00) Saskatchewan
US Mountain Standard Time	(GMT-07:00) Arizona
Mountain Standard Time (Mexico)	(GMT-07:00) Chihuahua La Paz Mazatlan - New
Mexico Standard Time 2	(GMT-07:00) Chihuahua La Paz Mazatlan - Old
Mountain Standard Time	(GMT-07:00) Mountain Time (US & Canada)
Pacific Standard Time	(GMT-08:00) Pacific Time (US & Canada)

Id	Display Name
Pacific Standard Time (Mexico)	(GMT-08:00) Tijuana Baja California
Alaskan Standard Time	(GMT-09:00) Alaska
Hawaiian Standard Time	(GMT-10:00) Hawaii
Samoa Standard Time	(GMT-11:00) Midway Island Samoa
Dateline Standard Time	(GMT-12:00) International Date Line West

C

CONNECTION STRING FORMATS

The format of the connections strings required in your configuration files depends on the configuration of your SQL Server. The following examples show some of the common connection string formats for MEM databases.

For more information on formatting connection strings, see SQL Server connection strings in the below link:

<http://www.connectionstrings.com/sql-server/>

Standard Security

```
Server=[ServerAddress];Database=[DataBaseName];User Id=[UserName];Password=[Password];
```

Trusted Connection

```
Server=[ServerAddress];Database=[DataBaseName];Trusted_Connection=True;
```

Asynchronous Processing

```
Server=ServerAddress;Database=[DataBaseName];Integrated Security=True;Asynchronous  
Processing= True;
```

Enable MARS

```
Server=myServerAddress;Database=myDataBase;Trusted_  
Connection=True;MultipleActiveResultSets=true;
```

Connection to a SQL Server Instance

```
Server=[ServerName]\[InstanceName]; Database=[DataBaseName];User Id=  
[UserName];Password=[Password];
```

Connect via an IP address

```
Data Source=[IPAddress],[PortNo];Network Library=[NetworkProtocol];Initial Catalog=  
[DataBaseName]; User Id=[UserName];Password=[Password];
```

WEBDAV SUPPORT MATERIAL

There are different ways to set up and secure a WebDAV server. The following material describes some of the basic procedures and best practices.

It is important that this activity involves IIS deployment experts to ensure the configuration is as appropriate and secure as possible for the hosting organization.

The minimum recommended WebDAV solution configuration is as follows:

- IIS 7+.
- SSL enabled.
- HTTP binding removed.
- Transfer root directory allowing read/write/modify access to a dedicated, unprivileged user.
- Transfer root directory mounted as a virtual directory.
- Transfer root directory containing minimal WebDAV authoring rules:
 - Allowable only for the unprivileged dedicated user.
 - Allowable only for the file types in use (e.x., *.csv, *.csv.gz, *.bin, *.bin.gz, *.pgd), and always including *.xfer.
 - Read/Write access only (no Source).
- All authentication methods disabled on Transfer root except for Basic.

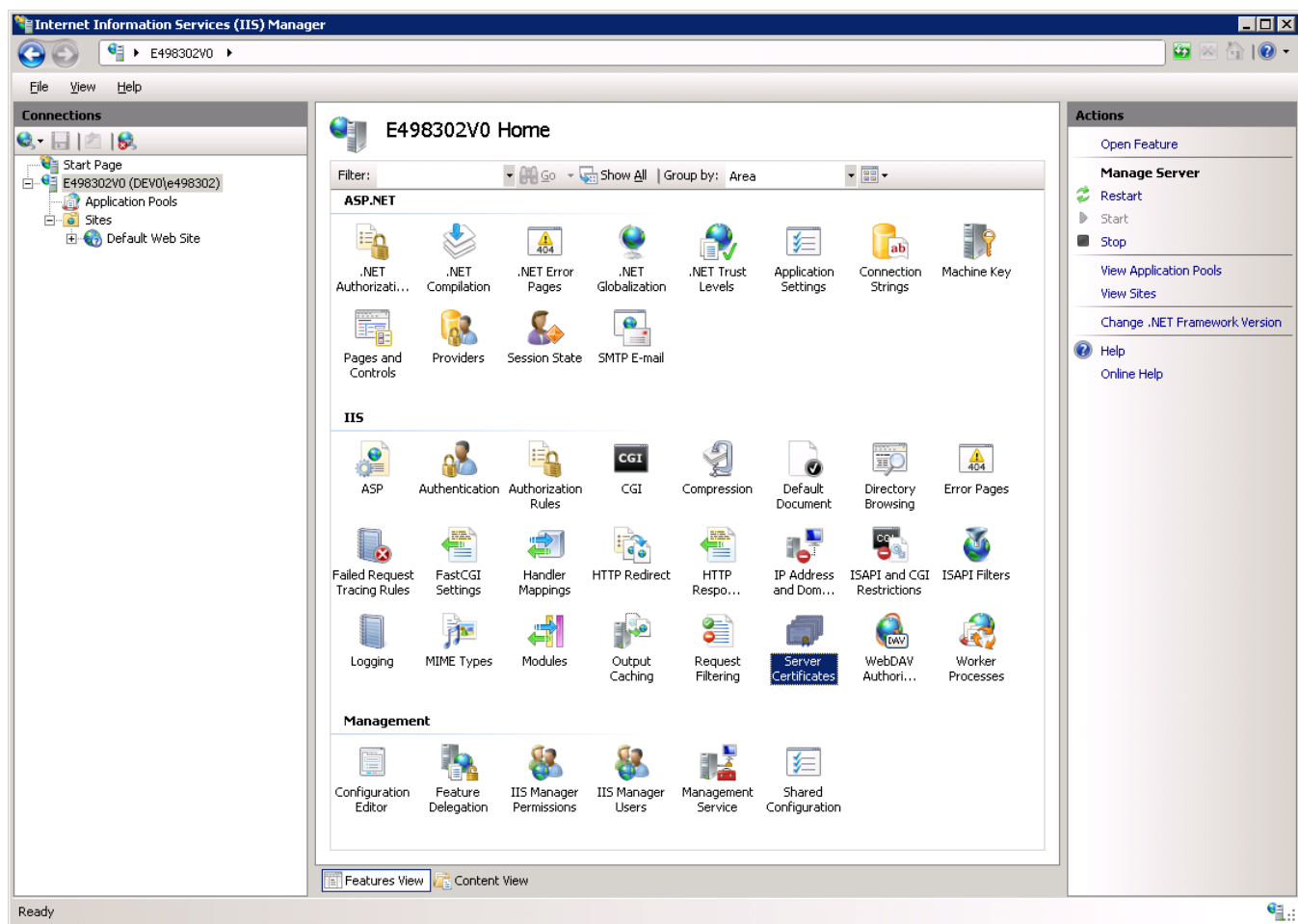
If both the Forwarding Agent and destination server share a common domain (or a trust exists between domains) and Kerberos is used for authentication within the domain, using Integrated Windows Authentication can be a good option.

If sufficient Public Key Infrastructure exists in the environment, the most secure option is likely to configure the Forwarding Agent to use a client X509 certificate which in turn is mapped to a local user on the IIS side. Certificate/client mapping in IIS is outside the scope of this document.

D.1 Configuring SSL

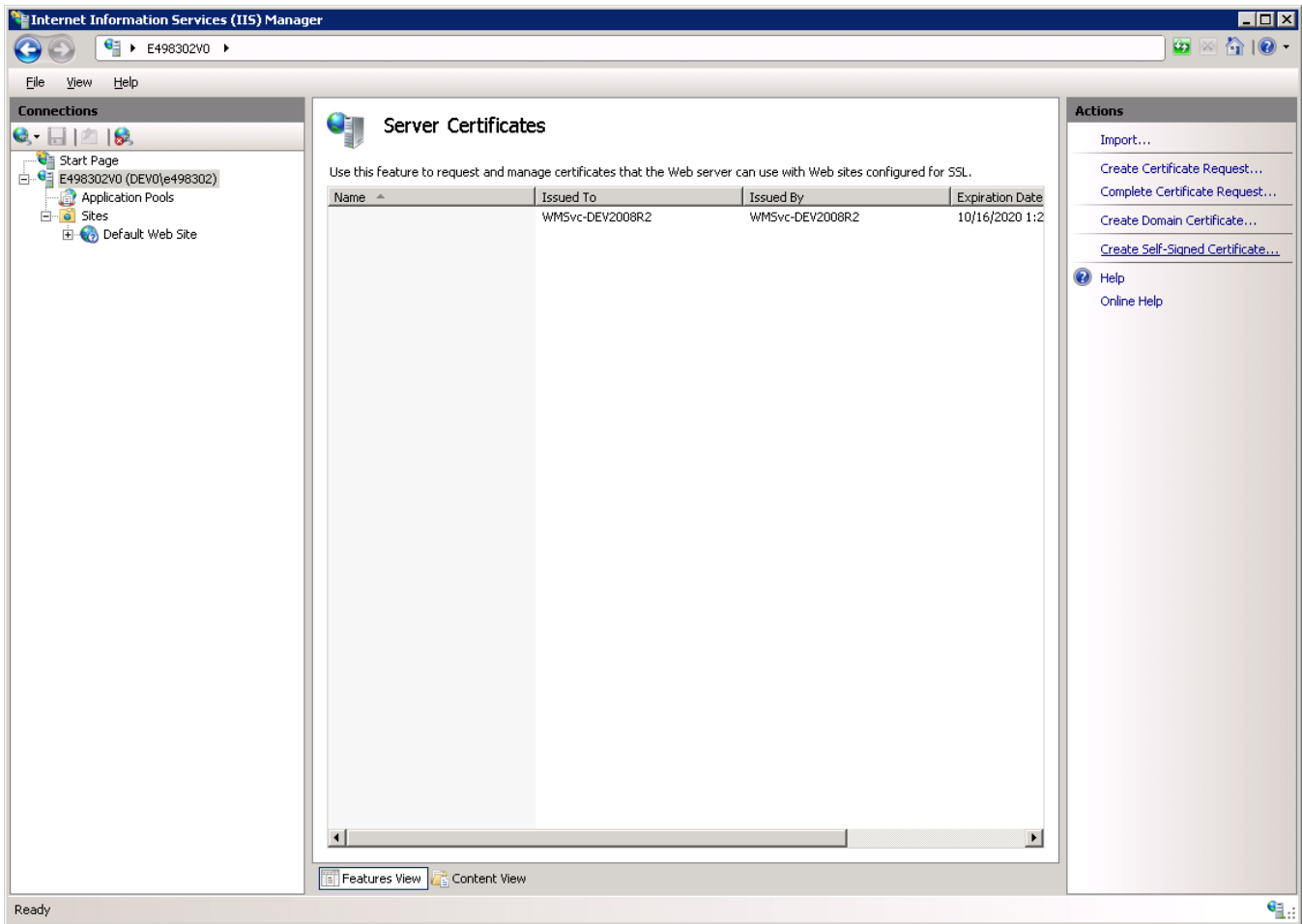
Assuming that IIS has already been installed on the transfer destination server, the following steps describe one way to configure and enable SSL. If standard procedures are available in the target environment, be sure to use those instead.

1. Open IIS Manager.
2. Select the machine, and double-click **Server Certificates** in the Features View.



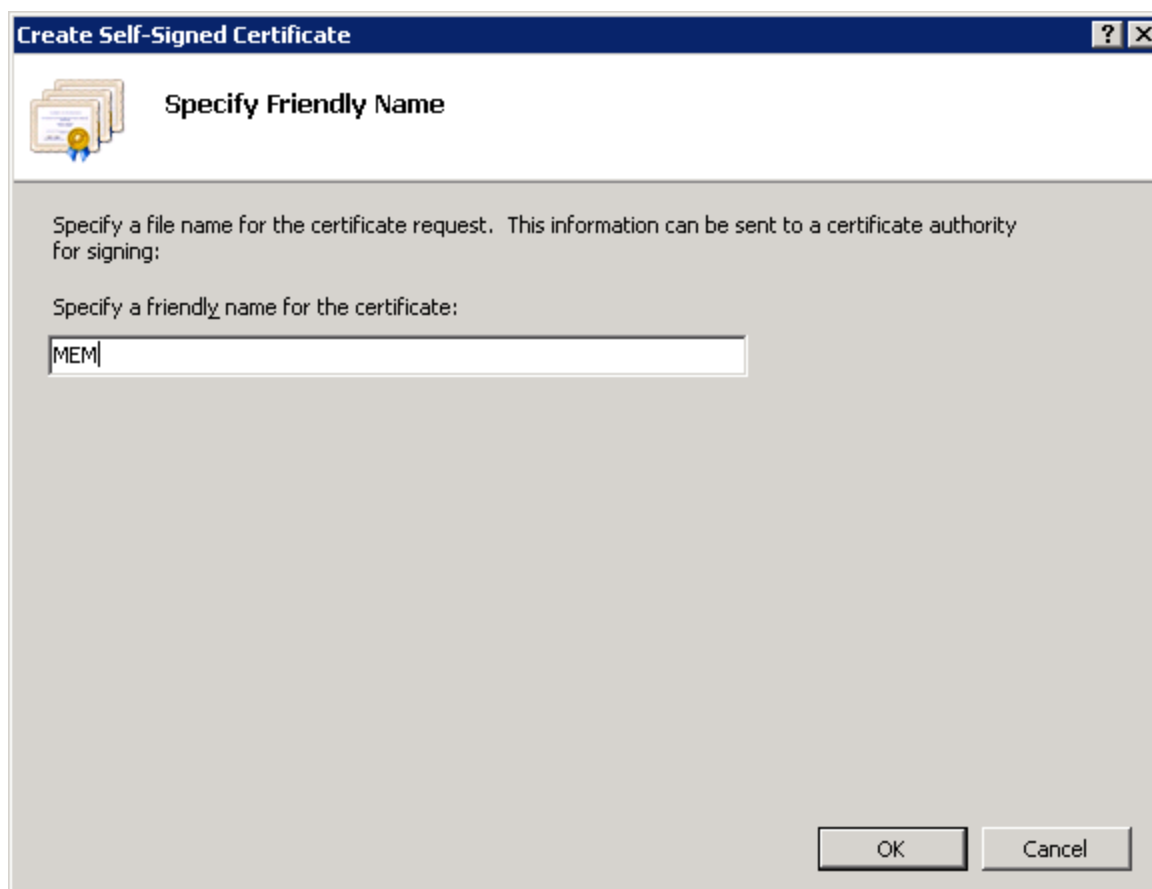
3. Determine if a valid server certificate already exists for the computer (in some cases certificates are automatically generated after installation).

A valid certificate entry is **not expired** and is issued to the **current name** of the computer.

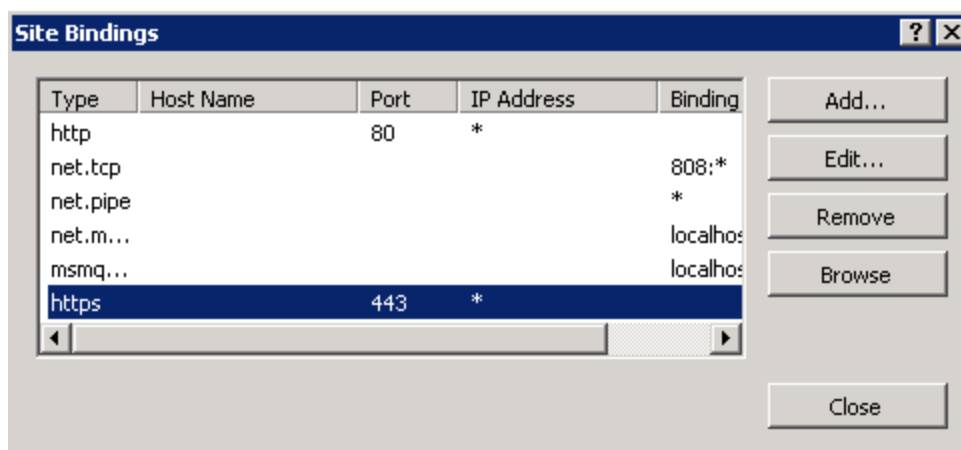


4. If no valid certificate is available, follow the below steps to create a new one:

- a. Click **Create Self-Signed Certificate**.
- b. Specify an appropriate **friendly-name** and click **OK**.

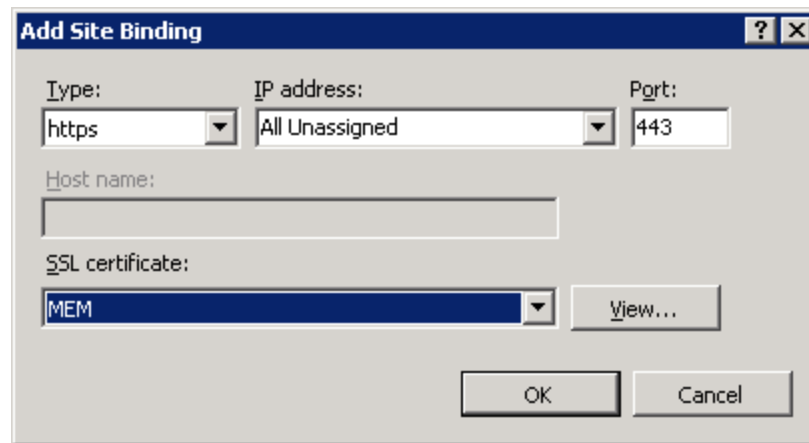


- c. In the main IIS Manager panel, right-click on the **Default Web Site** and select **Edit Bindings**.
- d. Click **Add** to add a new https binding:

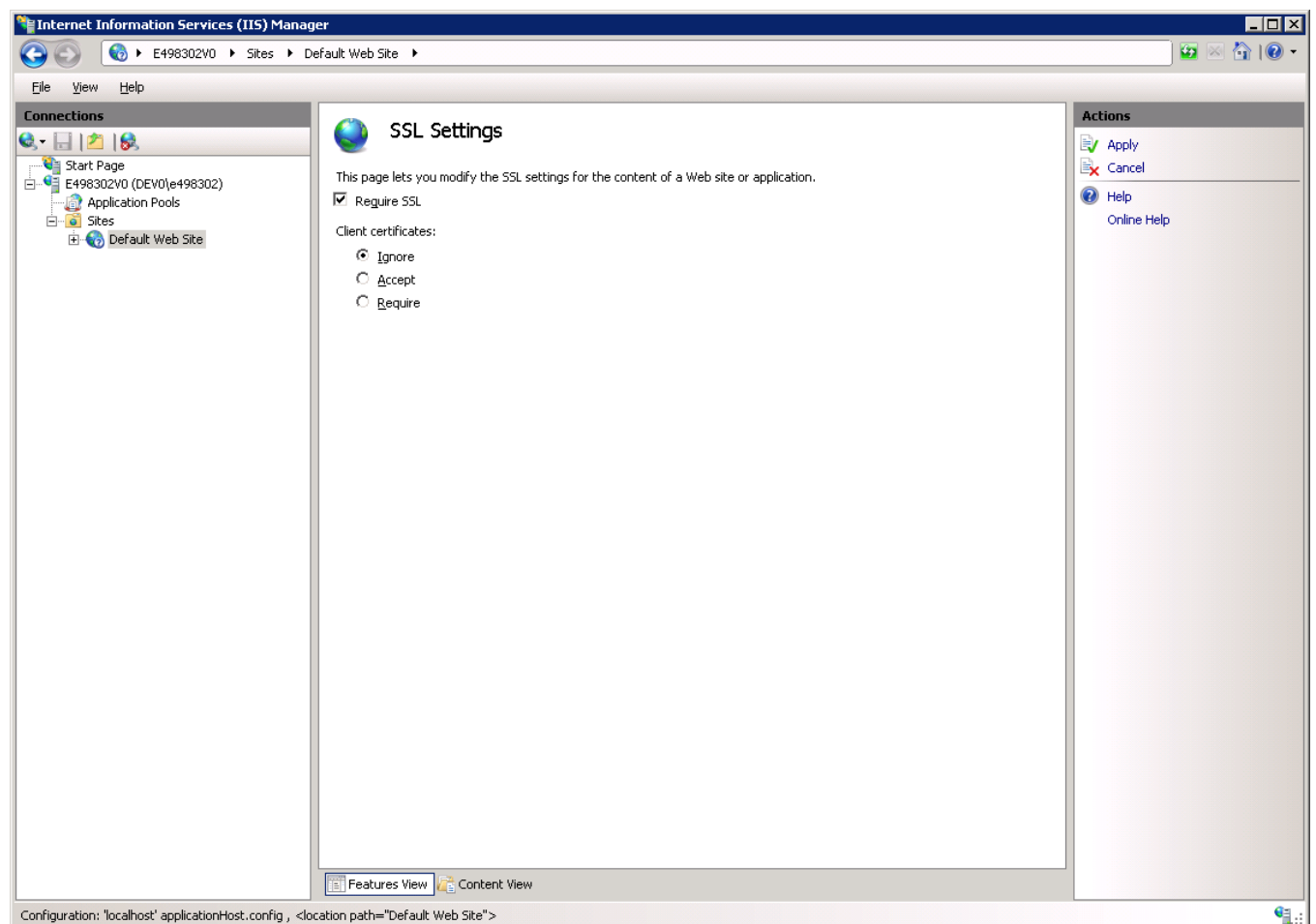


TIP
Remove the existing HTTP binding (Optional).

- i. Select **https** for the Type.
- ii. Keep the default port of 443.
- iii. Select the valid SSL certificate identified in the previous steps.



- e. Click **OK**.
- f. Click **Close**
- g. In the main IIS Manager panel, click the **Default Web Site** and double-click **SSL Settings** in the Features View.
- h. Select the **Require SSL** checkbox and click **Apply** action.



Test that SSL is set up correctly by opening a web browser and navigating to '**https://host**', where **host** is the fully-qualified host name for the server. The **IIS 7** welcome screen should be displayed without any certificate warnings.

D.2 Create the Transfer Root Directory and Transfer User

Create an unprivileged user who will have access only to the **incoming** directories on the destination server. This user should be a member only of the **Users** group and should be put through whatever lock-down procedure exists at site (disallowing RDP logon, etc).

Create a directory to serve as the root of all transferred files. Ensure that the sub-directory structure to hold all required equipment files is created under this root directory before using the Forwarding Agent.

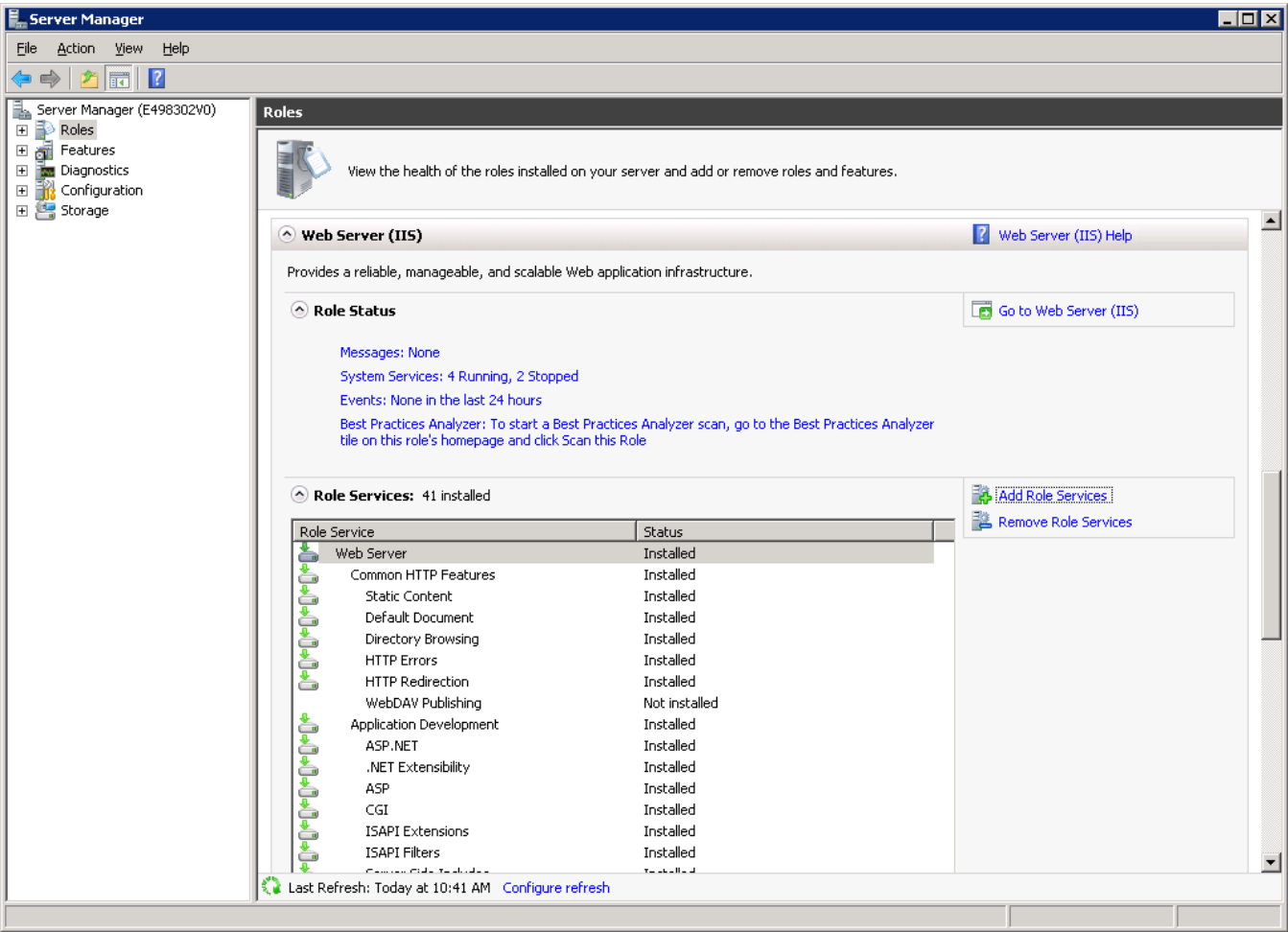
Specify the minimal set of permissions on the transfer root directory (which should propagate down the directory tree), which likely includes:

1. Administrators (full control).
2. The unprivileged transfer user (read/write/modify).
3. The Data Concentrator service user (read/write/modify).

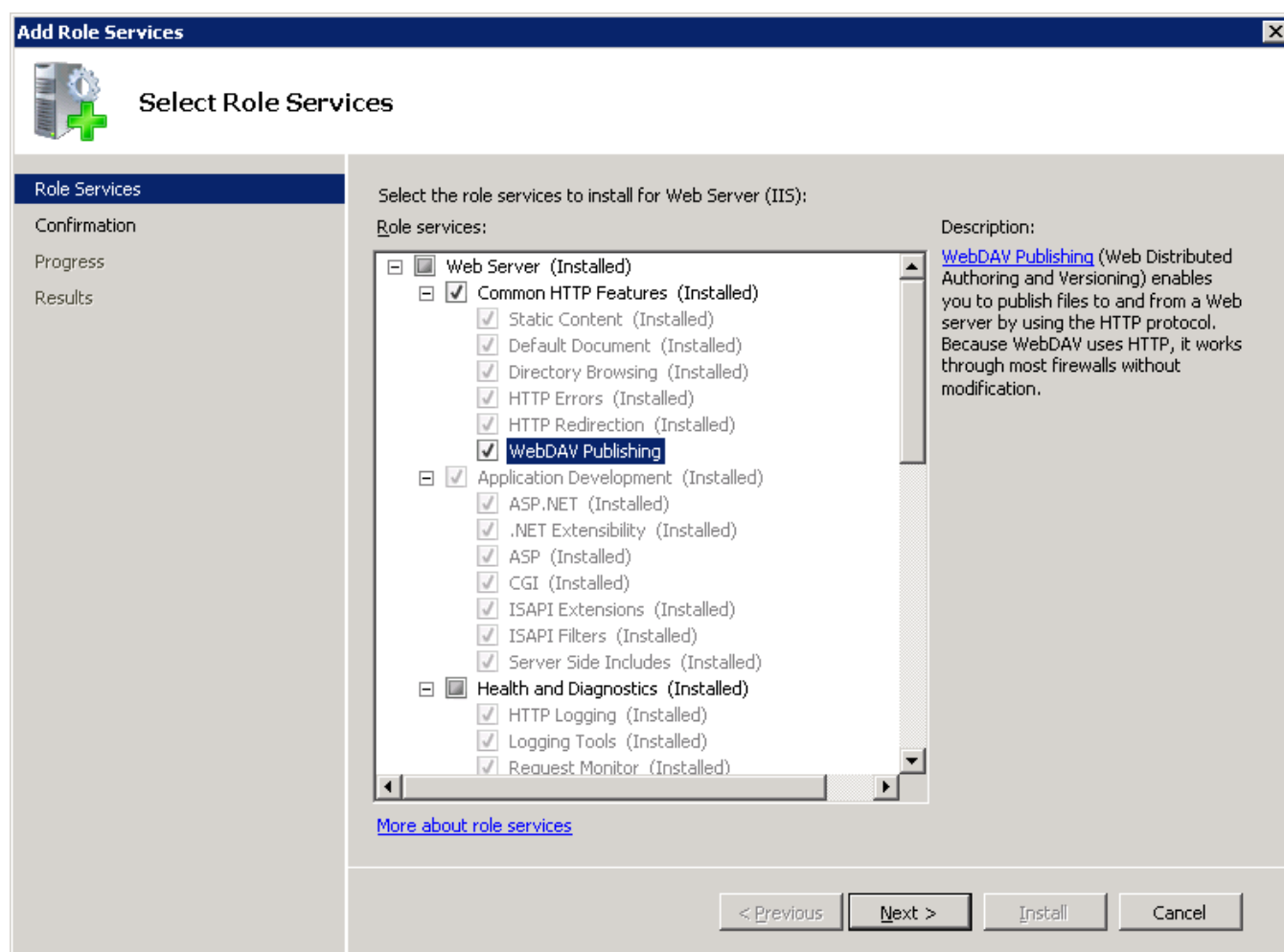
D.3 WebDAV Installation and Configuration

Having installed IIS and configured SSL, the next step is to enable the **WebDAV Publishing** IIS feature. Follow the below steps to enable WebDAV Publishing:

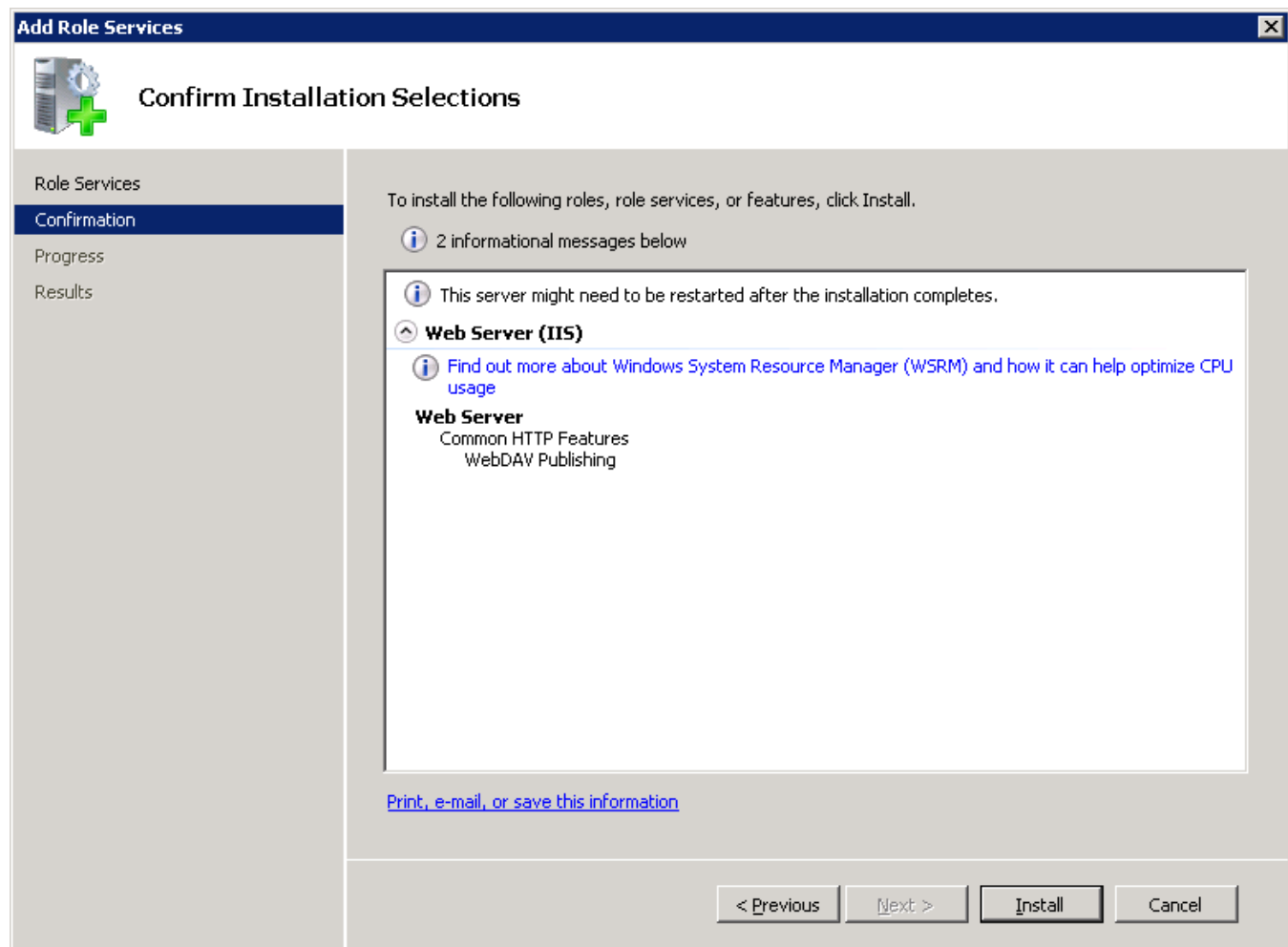
1. Open the **Server Manager** administrative tool.
2. Select **Roles** and find the **Web Server (IIS)** role.
Select **Add Role Services**.



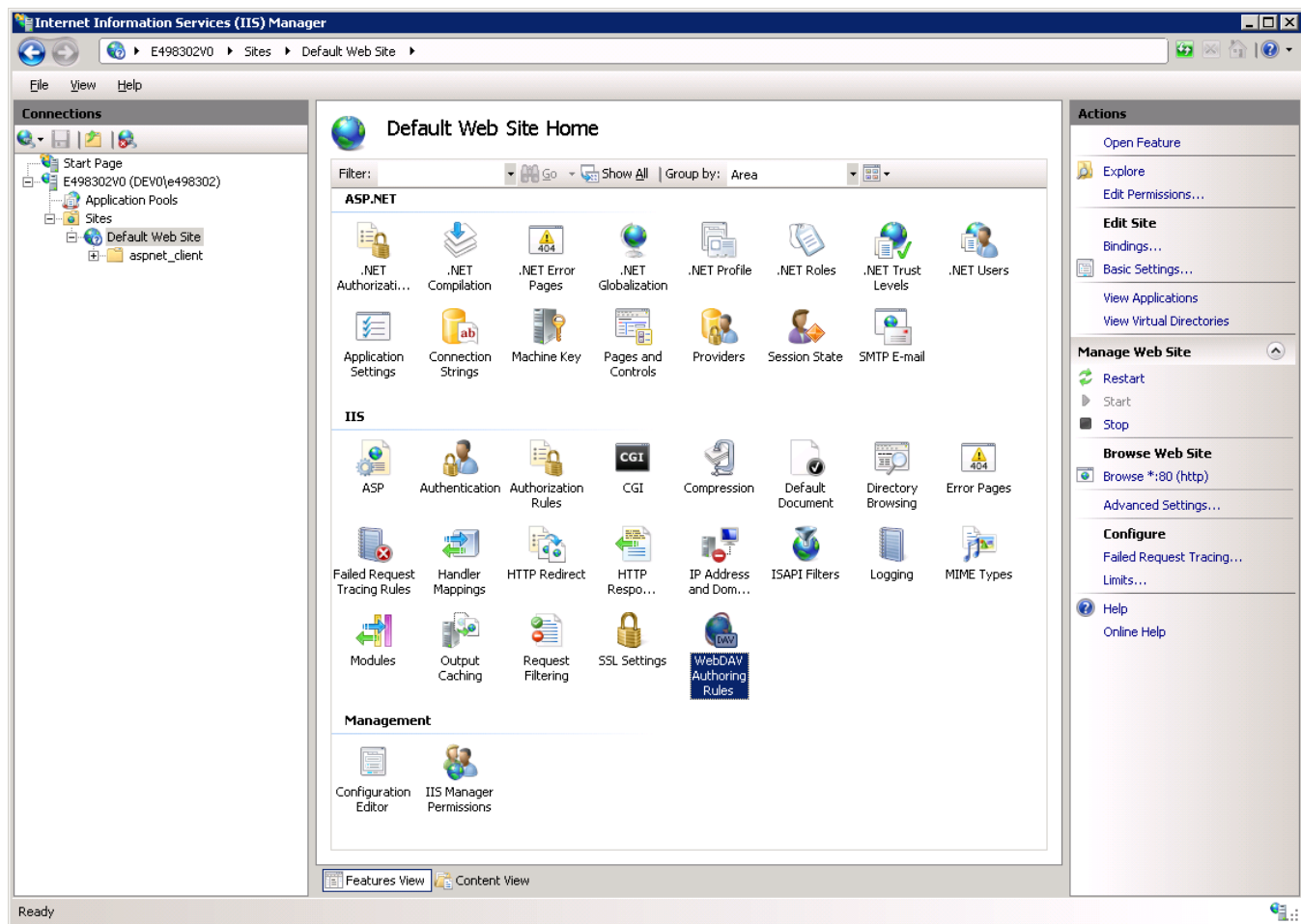
3. Select **WebDAV Publishing** and click **Next**.



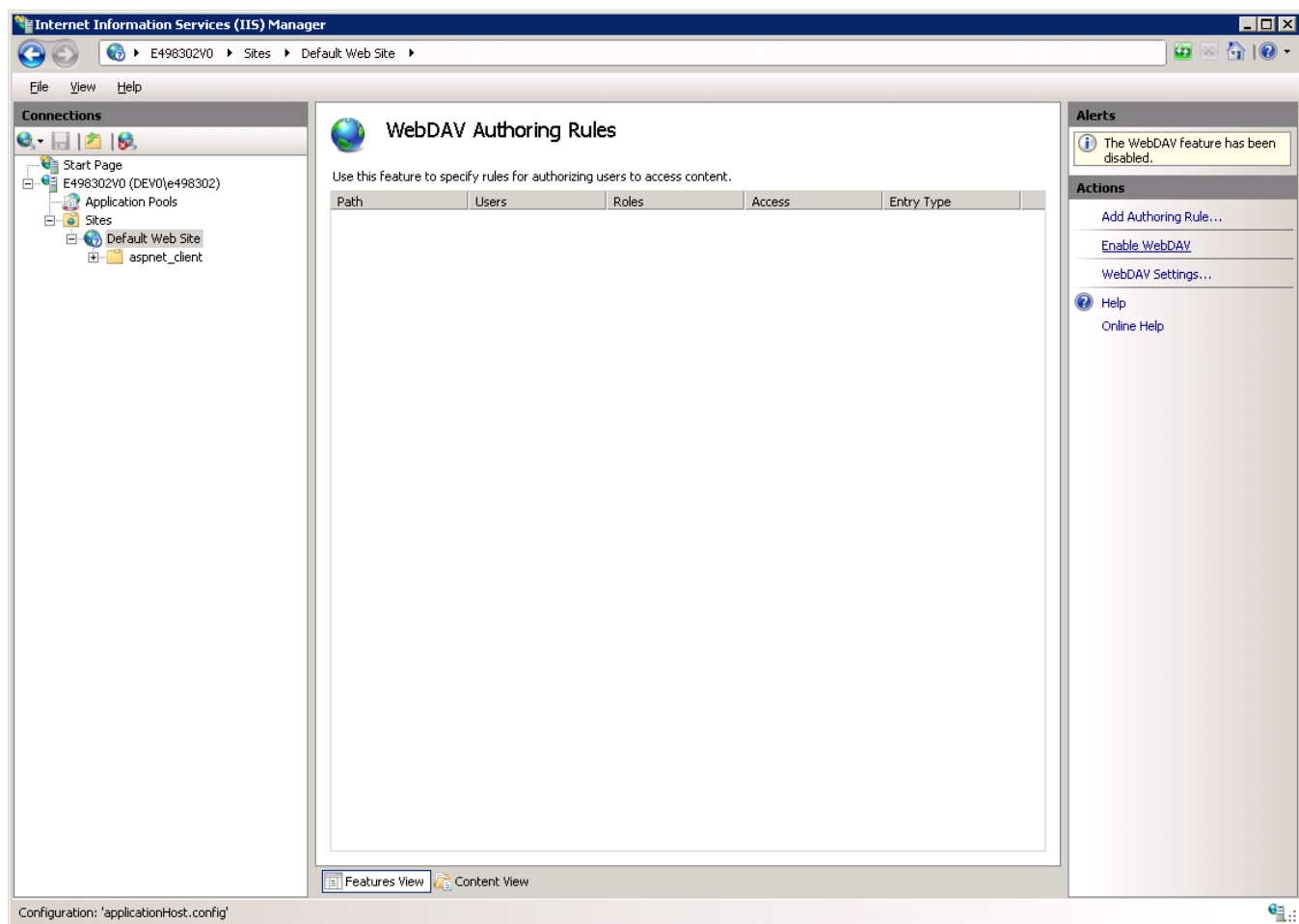
4. Click **Install**.



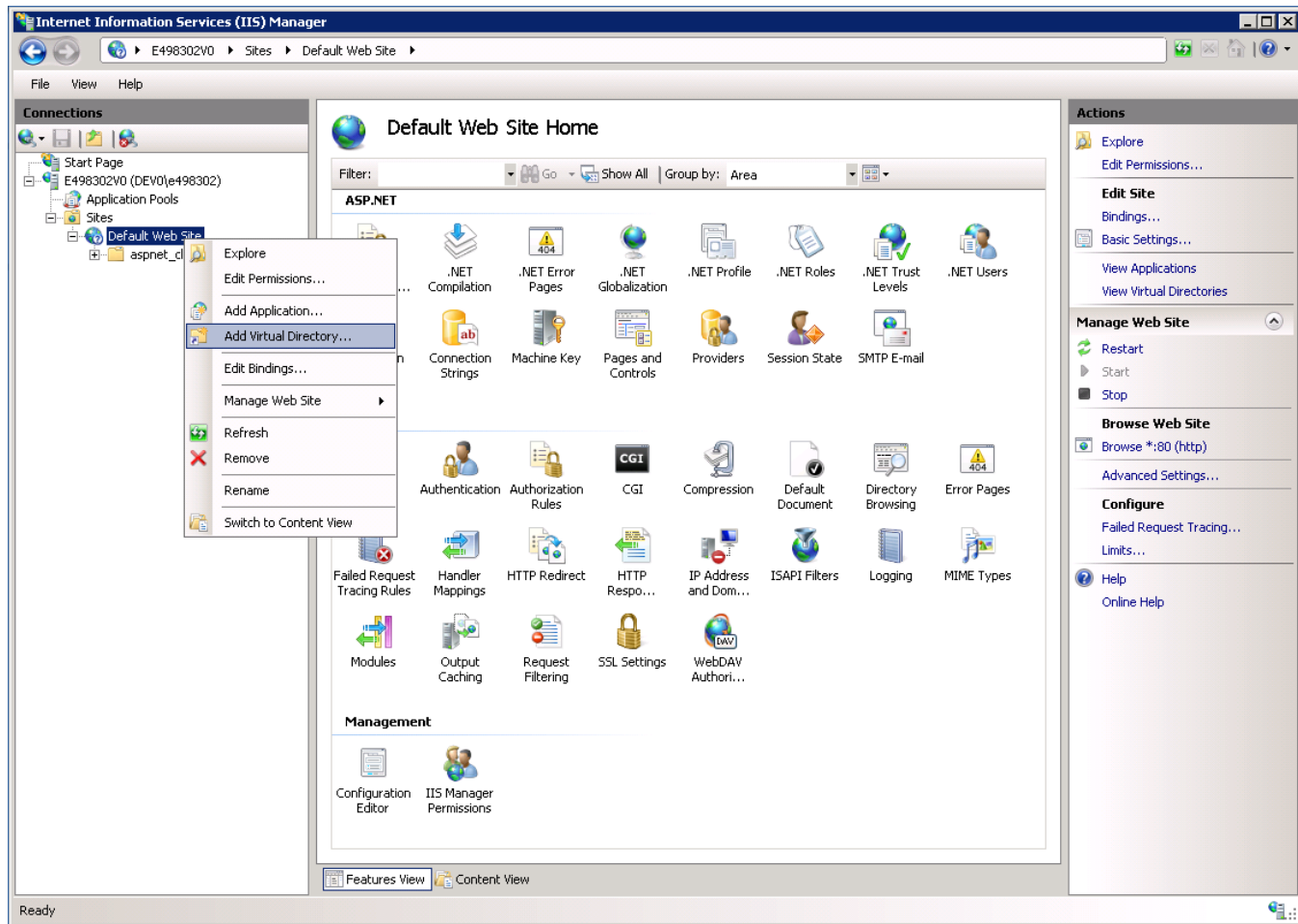
5. In IIS Manager, select **Default Web Site** and double-click **WebDAV Authoring Rules** in the Features View.



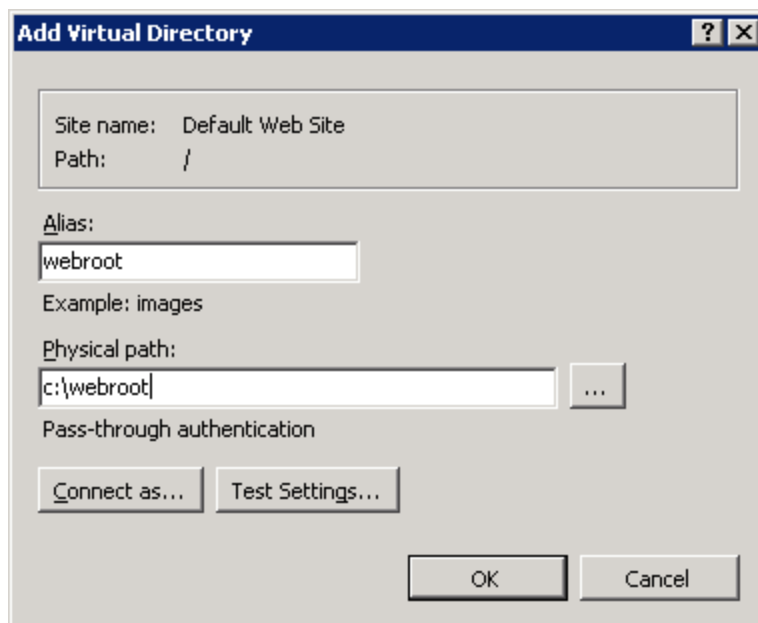
6. Click **Enable WebDAV** in the right-hand **Actions** panel.



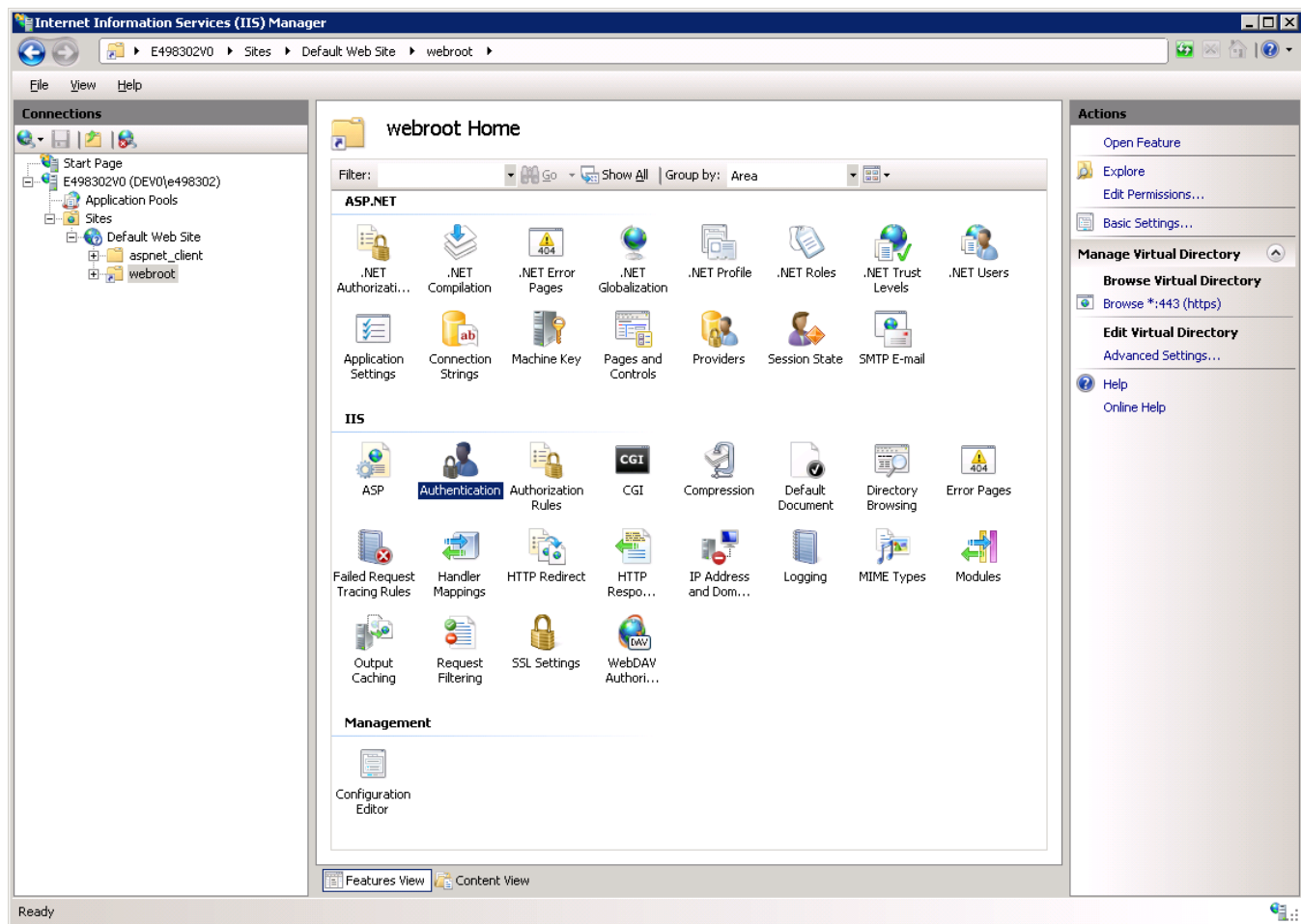
7. In the main IIS Manager panel, right-click **Default Web Site** and select **Add Virtual Directory**.



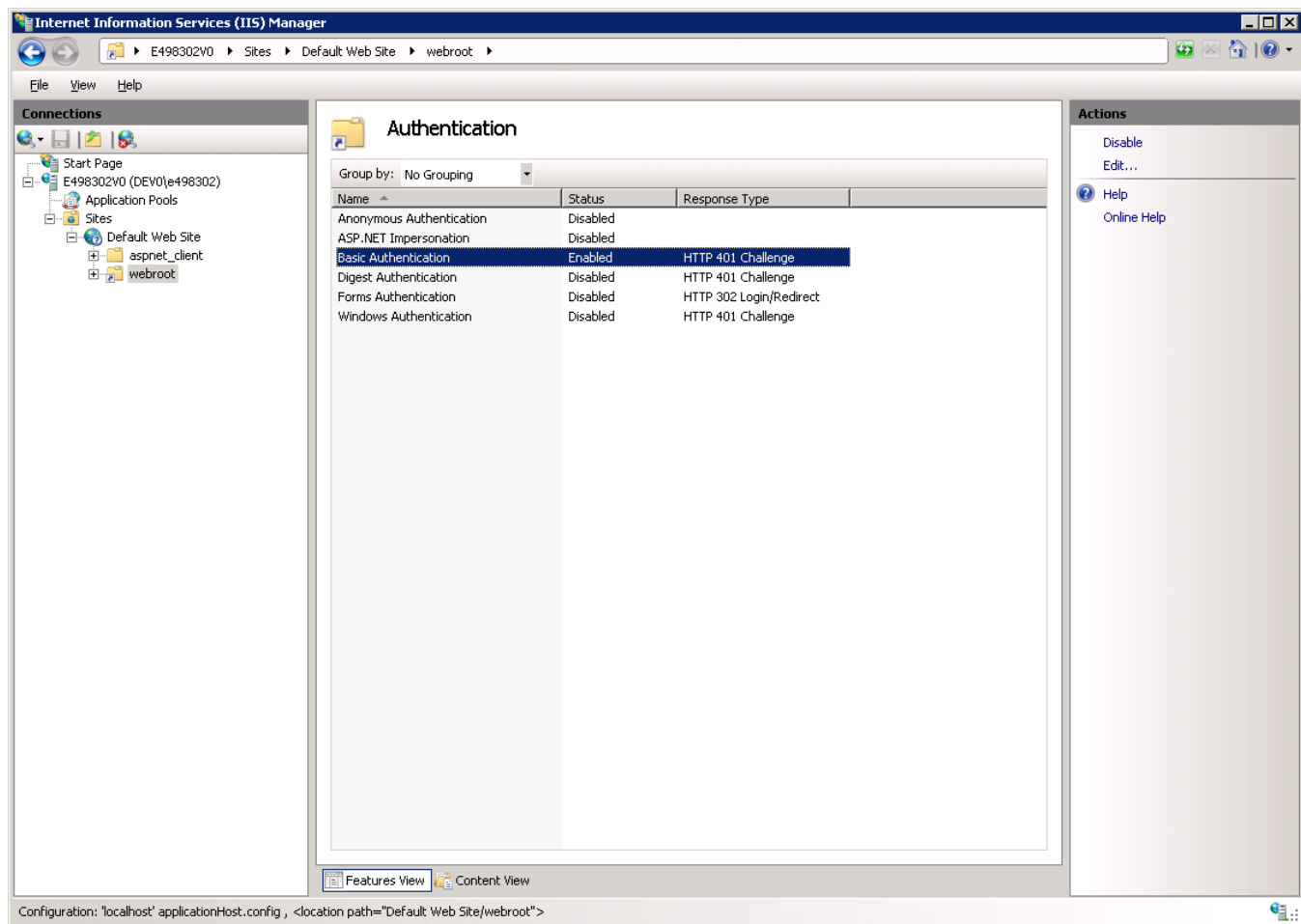
8. Specify an alias (e.x., 'webroot') pointing to the transfer root directory defined above.



9. Select the new virtual directory (e.x., 'webroot') and double-click **Authentication** in the Features View.



10. Disable all authentication methods except for the method chosen for your environment (in this example, Basic Authentication).



11. In the main IIS Manager panel, select the new virtual directory (e.x., 'webroot') and double-click **WebDAV Authoring Rules**.
12. Click **Add Authoring Rule** in the **Actions** panel and update the following fields:
 - a. **Allow access to:** All content.

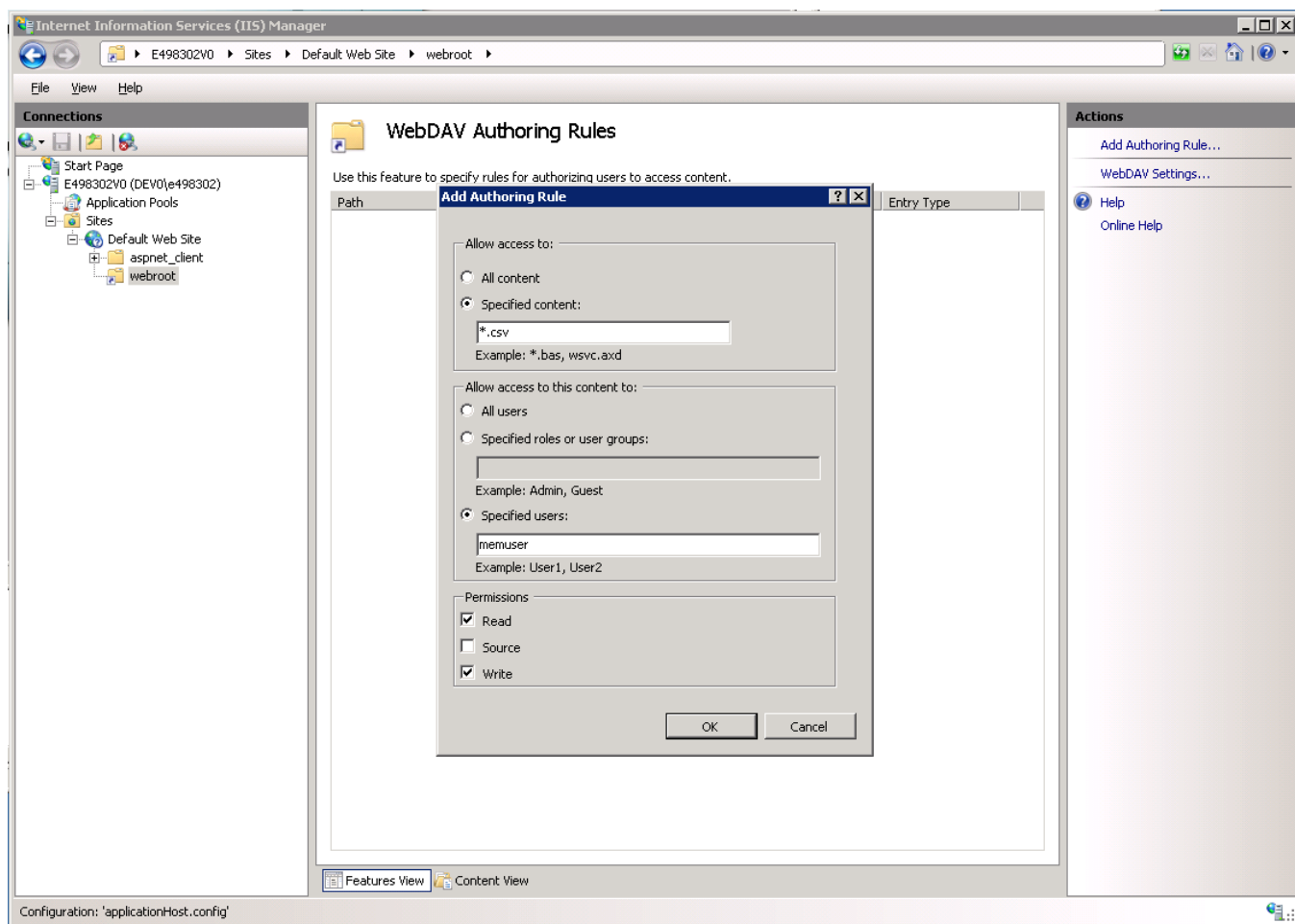
NOTE

A more secure option is to select **Specified content** and specify the exact wildcards and extensions that are expected to be used in the MEM system (and multiple extensions must be specified in multiple WebDAV Authoring rules).

These extensions include:

- *.xfer (used as intermediate files by the Forwarding Agent)
- *.csv (if using uncompressed sensor files)
- *.pgd (if using uncompressed alarm files)
- *.log (if using uncompressed log files)
- *.gz (if using compressed sensor/alarm/log files)

- b. **Allow access to this content to:** Specified users.
Enter the unprivileged user to be used by the Forwarding Agent.
- c. **Permissions:** Read/Write.

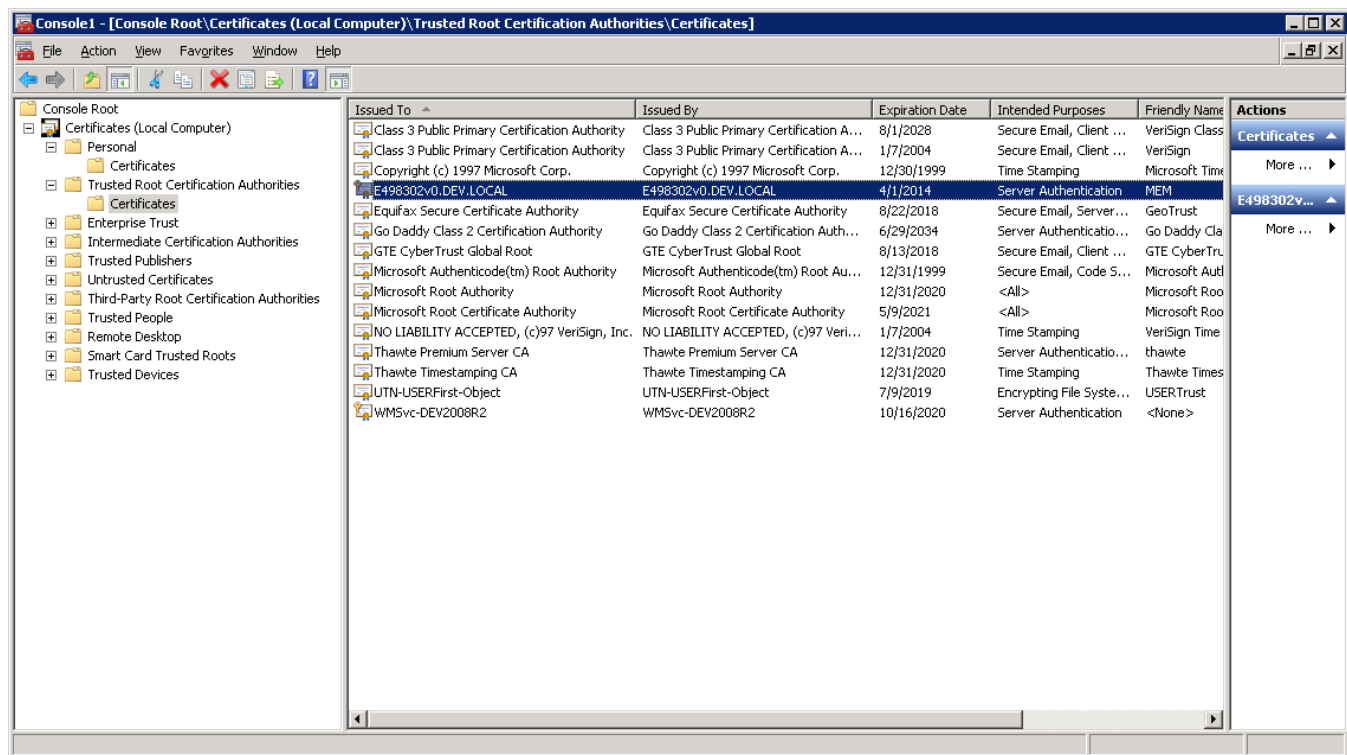


D.4 Certificate Exchange

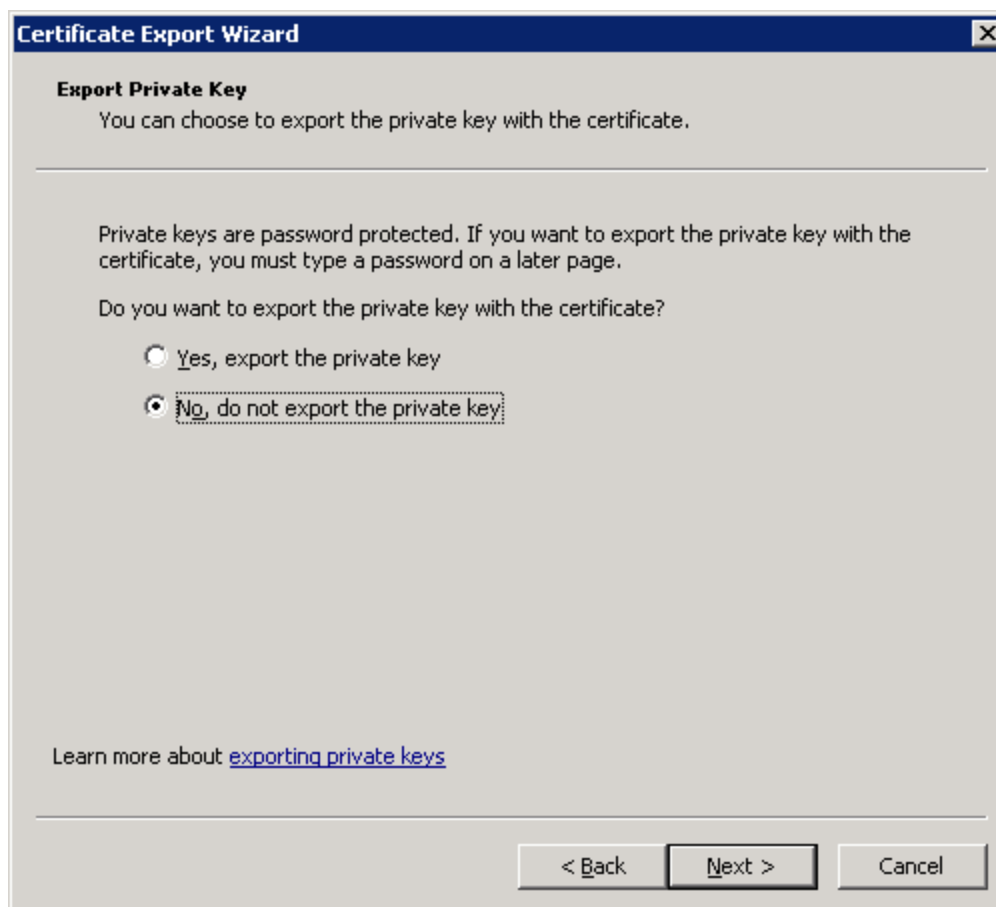
If the destination server's SSL certificate was not issued by a common trusted CA (Certification Authority), this certificate must be present on the Forwarding Agent machine before the Forwarding Agent will trust the server identity.

Perform the following steps in the destination server:

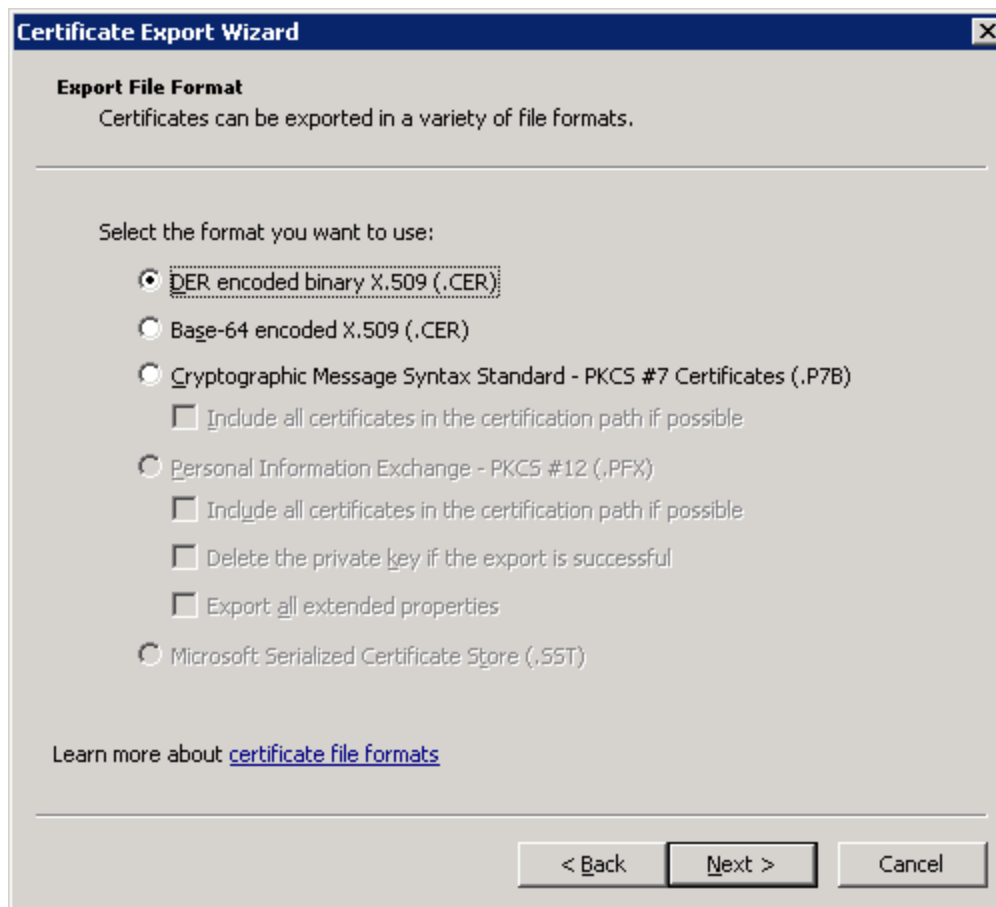
1. Launch the Microsoft Management Console (Start / Run / mmc).
2. Click File / Add/Remove Snap-In.
3. Select Certificates, then click **Add**.
4. Select Computer account for the This snap-in will always manage certificates for option. Click **Next**.
5. Select Local computer for the This snap-in will always manage option. Click **Finish**.
6. Click **OK** on the Add or Remove Snap-ins dialog.
7. Drill-down to Console Root / Certificates (Local Computer) / Personal / Certificates, and select the appropriate server certificate.



8. Right-click the certificate and select All Tasks / Export.
Select **No, do not export the private key** and click **Next**.



9. Select **DER encoded binary X.509 (.CER)** and click **Next**.



10. Enter a filename with a **.cer** extension and click **Next**,
11. Click **Finish**.

Perform the following steps in the Forwarding Agent server:

1. Copy the destination server certificate exported in the previous step.
2. Launch the Microsoft Management Console (Start / Run / mmc)
3. Click File / Add/Remove Snap-In
4. Select Certificates, then click **Add**
5. Select Computer account for the This snap-in will always manage certificates for option. Click **Next**.
6. Select Local computer for the This snap-in will always manage option and click **Finish**.
7. Click **OK** in the Add or Remove Snap-ins dialog.
8. Drill-down to Console Root / Certificates (Local Computer) / Trusted Root Certification Authorities / Certificates. Right-click and select All Tasks / Import.
9. Select the target server certificate and click Next.
10. The Place all certificates in the following store / Trusted Root Certification Authorities and click **Next**,
11. Click **Finish**.

D.5 Configuring a Web Proxy

If the site environment requires HTTP traffic to go through a web proxy, this can be configured in the Forwarding Agent configuration file. The default configuration file contains a commented-out <defaultProxy> sample section in the <system.net> section. Uncomment this section so that it resembles:

```

<system.net>

<defaultProxy enabled="true" useDefaultCredentials="true">

<proxy
  usesystemdefault="true"
  proxyaddress="http://192.168.1.10:3128"
  bypassonlocal="true"
/>

</defaultProxy>

</system.net>

```

Set the correct proxy address and port in the **proxyaddress** attribute.

Setting **useDefaultCredentials** to **True** will authenticate with the Forwarding Agent service's user if required. Setting this attribute to **False** will disable authentication.

Refer <http://msdn.microsoft.com/en-us/library/kd3cf2ex.aspx> for more information.

D.6 Configuring an Authenticated Web Proxy

To configure a web proxy with explicit credentials, configure the **<defaultProxy>** section as follows:

```

<system.net>

<defaultProxy enabled="true" useDefaultCredentials="false">

<module type="Matrikon.Datalogger.ForwardAgent.Service.AuthenticatedProxy,
Matrikon.Datalogger.ForwardAgent.Service" />

</defaultProxy>

</system.net>

```

Set the correct proxy address and port in the **proxyaddress** attribute.

Setting **useDefaultCredentials** to **True** will authenticate with the Forwarding Agent service's user if required. Setting this attribute to **False** will disable authentication.

Refer <http://msdn.microsoft.com/en-us/library/kd3cf2ex.aspx> for more information

Then, configure the proxy address and credentials in the **<authenticatedProxy>** section:

```

<authenticatedProxy proxyaddress="http://192.168.1.10:3128" user="user1"
password="password1" domain="" />

```

The user, password and domain fields can reference protected settings, as described in section [Protected Settings](#).

D.7 Configuring Maximum Web Connections

Section [WebDAV](#) describes the **maxConcurrency** setting on a WebDAV transfer connection. This setting internally controls how many simultaneous transfers are attempted on that connection, but the system imposes an additional restriction that is controlled through the **<connectionManagement>** section.

To increase the system limit from the default (typically, 2) to 4 simultaneous connections, use the following:

```
<system.net>
<connectionManagement>
<add address="*" maxconnection="4"/>
</connectionManagement>
</system.net>
```

SECURITY CONSIDERATIONS

To increase the security of your Data Concentrator installation, you should make some changes to the default settings. These changes will configure the Data Concentrator to use connections with the lowest level of security access possible. This will also protect the configured credentials.

Follow the below steps to increase the security of Data Concentrator:

1. Execute the following script to create the required database roles (scripts provided) and then add the appropriate login/users to these roles:

```
MEM\Database Scripts\DatabaseRoles\DataConcentrator_role.sql
```

2. Use the installation wizard to protect the Data Concentrator configuration files, as described below.

To Protect the Data Concentrator Configuration Files

1. Navigate to the following folder in the installation media and double-click the **Honeywell APM Data Concentrator.msi** file:

<Installation Media>:\Utilities\DataConcentrator\DataConcentrator\

2. The installation wizard appears.

NOTE

Any components installed on the system will have the option to protect the configuration settings.

3. Select **Data Concentrator Configuration Protection**, and click **Next**.
4. Click **Install**.

The configuration files are protected.

E.1 Configuring DC for APM

If you are using the Data Concentrator with APM (refer [Asset Performance Management](#)), you should make the following changes:

1. Exchange certificates between the Data Concentrator computer and the APM Application Server.
2. Update the Data Concentrator services user's security permissions.

E.1.1 Exchanging Certificates

Perform the export and import procedures on both the Data Concentrator computer and the APM Application Server.

To Export the Certificates

1. Open the Console Root (**Start > Search > mmc**).
2. Follow the below steps to add the Certificates Snap-in:
 - a. From the **File** menu, select **Add/Remove Snap-in**.
 - b. Double-click **Certificates**.
 - c. Select **Computer Account**, and click **Next**.
 - d. Click **Finish**.
 - e. Click **OK**.
3. From the tree, expand **Certificates**, and select **IntuitionSign**.
4. In the certificate list, right-click the certificate, point to **All Tasks**, and select **Export**.
5. Complete the steps in the Certificate Export Wizard, and save the certificate in a location that is accessible or can be transferred to the other computer.

To Import the Certificates

1. In the Console Root, on the tree, expand **Certificates**, and select **IntuitionVerify**.
2. In the **Actions** sidebar, click **More Actions**, point to **All Tasks**, and select **Import**.
3. Complete the Import Certificate Wizard for all of the following:
 - a. **In the APM Application Server**: The certificate exported from the Data Concentrator computer.
 - b. **In the Data Concentrator computer**: The certificate exported from the APM Application Server.
4. For both certificates, update the friendly names.

To Update the Friendly Names

1. Double-click the imported certificate in the **IntuitionVerify** folder.
2. Follow the below steps in the **Certificate** dialog box:
 - a. Click the **Details** tab.
 - b. Click the **Friendly name** box.
 - c. Copy the friendly name (starts with CN=...) to the clipboard.
 - d. Click **OK** to close the Certificate dialog box.
3. Right-click the same certificate, and select **Properties**.
4. Perform the following steps in the **Properties** dialog box:
 - a. In the Friendly name box, paste the Friendly name.
 - b. Click **OK** to close the **Properties** dialog box.

E.1.2 Updating Security Permissions

Add Services User to the Windows Reliability Engineers Group

To ensure the Data Concentrator can access Asset Performance Management, the account the Data Concentrator is running as must be added to the Reliability Engineers group.

Follow the below steps to add the account to Reliability Engineers group:

1. Open the Server Manager (**Start > Server Manager**).
2. On the server tree, choose **Configuration > Local Users and Groups > Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter the name of the user that runs the Data Concentrator service.
6. Click **OK**.
The services user appears in the group list.
7. Click **OK** to close the Reliability Engineers Properties dialog box.

To Give Permission to the MachineKeys Folder

To ensure the Data Concentrator has access to the required files, the account the Data Concentrator is running as must be given permission to access the MachineKeys folder.

Perform the following steps on the Asset Performance Management Web, Application, and Calculation servers.

1. Locate the following folder in the server:
<install drive>:\ProgramData\Microsoft\Crypto\RSA\MachineKeys
2. Right-click the **MachineKeys** folder, and select **Properties**.
3. In the **Properties** dialog box, click the **Security** tab.
4. Click **Edit**.
5. Click **Add**.
6. Add the user the account that runs the Data Concentrator service, and click **OK**.
7. Select the Data Concentrator service user and add the following permissions:
 - a. **Read & Execute**
 - b. **List folder contents**
 - c. **Read**
8. Click **OK** in the Permissions dialog box.
9. Click **OK** in the Properties dialog box.

F.1 Custom Counter File Memory Error

F.1.1 Symptom

The following error appears in the log files:

WARN Collector - GetLastCounterValue Error: Custom counters file view is out of memory.

Refer <https://msdn.microsoft.com/en-us/library/ms229387%28v=vs.110%29.aspx?f=255&MSPPErr=-2147217396> for information on adjusting machine.config performance counter memory setting.

F.1.2 Resolution

The ASP .Net 4.0 configuration file needs to be changed to increase the value of the filemappingsize.

For more information, refer the following Microsoft article:

<https://msdn.microsoft.com/en-us/library/ms229387%28v=vs.110%29.aspx?f=255&MSPPErr=-2147217396>

1. Open the ASP .Net machine.config file for editing.

For example, on an x64 computer, the configuration file should be at the following location:

C:\Windows\Microsoft.NET\Framework64\v4.0.30319\Config

2. Add the following code, with an increased memory limit in bytes (default is 524,288 bytes), to the **<configuration>** section:

```
<system.diagnostics>
<performanceCounters filemappingsize="[memorylimit]" />
</system.diagnostics>
```

3. Save and close the file.
4. Restart the computer.

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